Zn incubation study

Raw recordings to reviewed events and publication figures

Follow one page at a time. Orange numbers on each screenshot match the numbered actions beside it. Numbers restart on the next page.

01

Prepare

Identify each ABF and apply the exact recording recipe.

02

Review

Check detections; optionally edit individual events or whole clusters.

03

Analyze & style

Recompute, make the complete figure set, style it and save.

Choose your route. Keep the automatic event decisions to reproduce the saved reference. For a curated analysis, use a separate workspace copy and rerun the affected downstream stages.

The Zn reference uses the supplied student. Native model-free detection is available for a separately documented analysis.

Launcher and shared curation views are refreshed for build 40.15.59. Unchanged reference illustrations retain their capture dates. Examples show saved or explicitly labeled fixture data; this documentation update did not rerun the scientific studies.

Find the next step

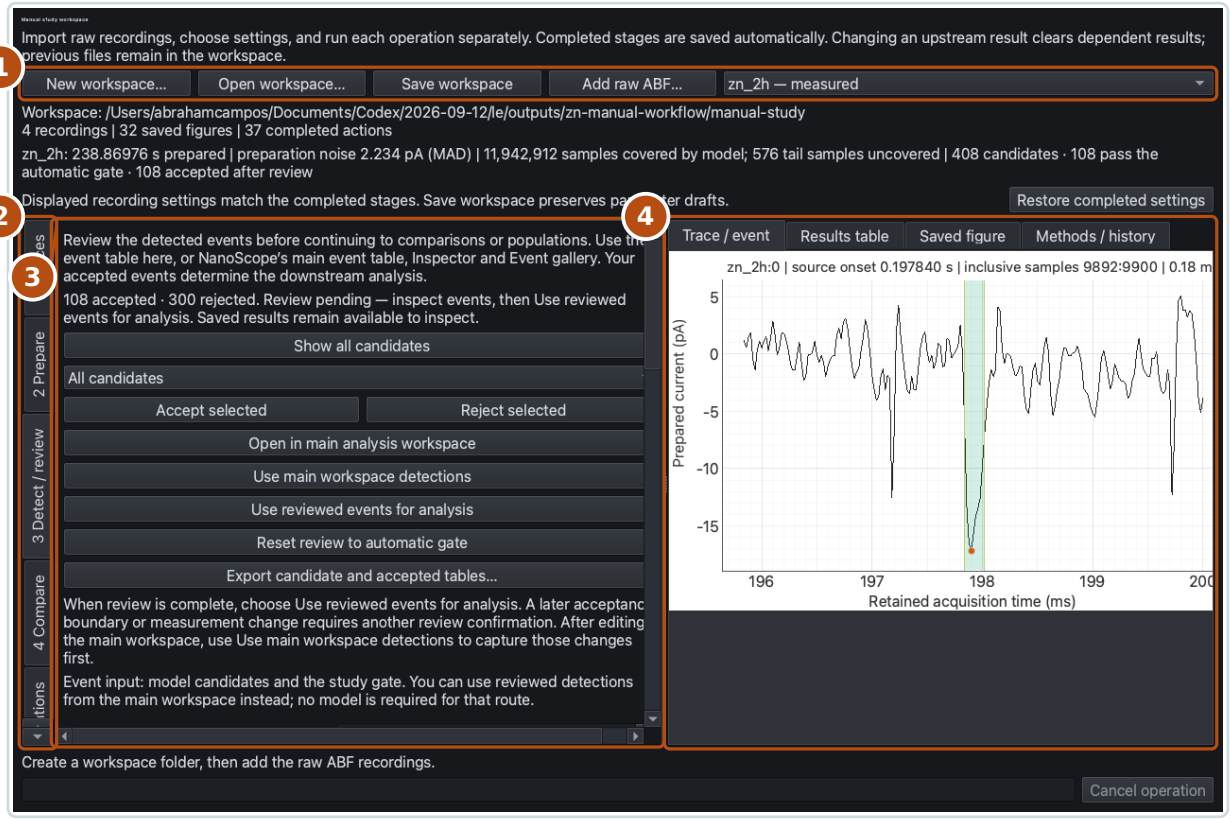
Use the page IDs in the text to find the corresponding walkthrough or recipe.

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First, see the workspace overview on page 3. Recipes and checkpoints follow the walkthrough: exact file names, scientific settings, counts and the figure inventory. Use the PDF bookmarks to jump directly to any walkthrough or recipe.

Find your way around the workspace

Study controls and result views. Use the following close-ups for the current launcher and shared event-curation steps.



Reference capture, 21 Sep 2026. NanoScope reference controls and saved results.

1 Choose the recording

The top toolbar creates, opens and saves the workspace; the selector chooses the active recording.

2 Choose the stage

The six tabs run from Sources to Figures. If a tab is hidden, use the small arrow at the bottom of the tab strip.

3 Set and run

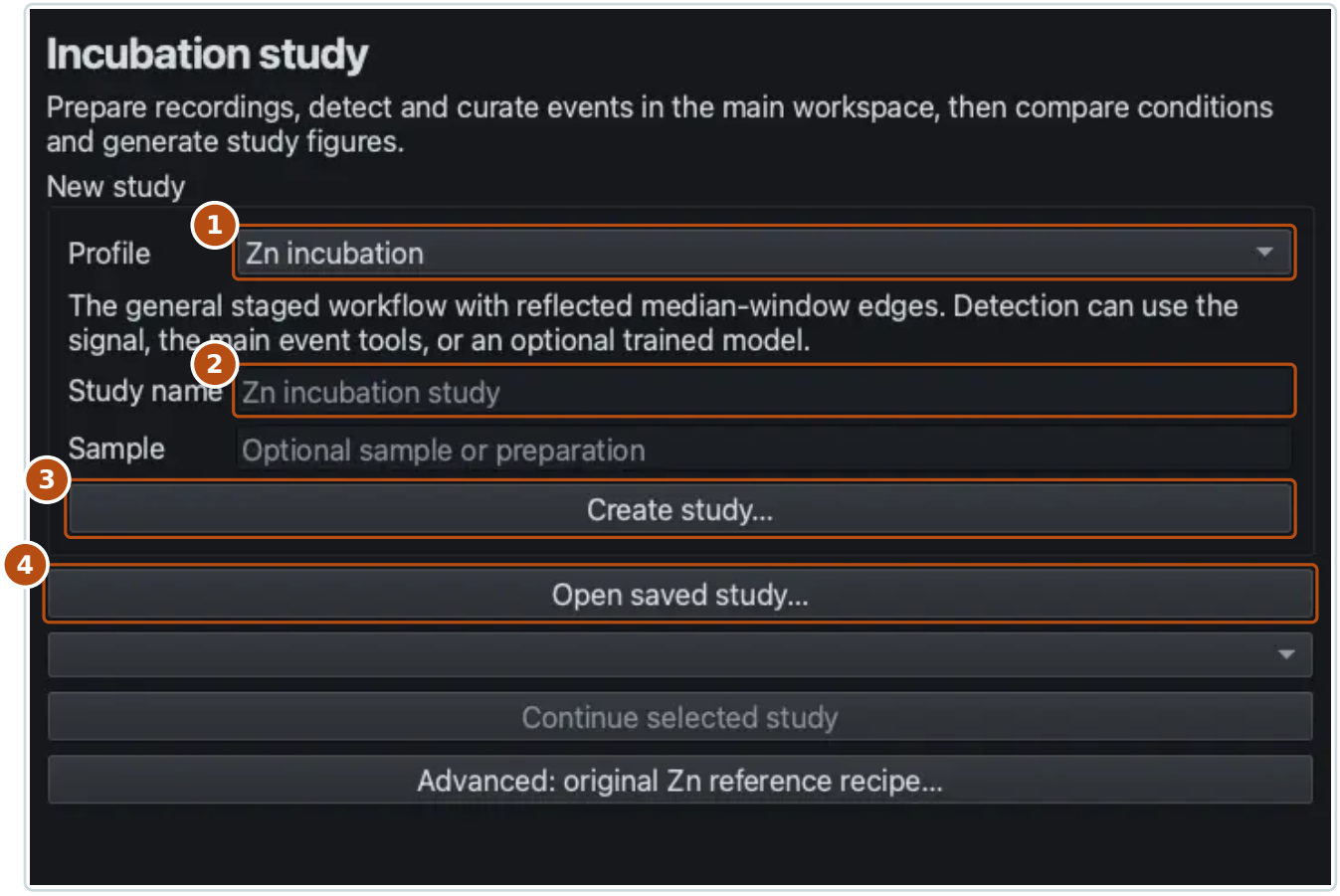
Enter values and click the stage button here. Scroll inside this panel to reach the lower controls.

4 Inspect the result

Switch among the trace, tables, figures and methods here. These tabs display the saved results.

Create or reopen an incubation study

Unified incubation workflow | NanoScope 40.15.59



Current unified study launcher · zn

DO THIS IN NANOSCOPE

1. **Choose the profile**
   Open Analysis > Incubation study... in the main app. Choose Zn incubation. Use General incubation for another sample or condition.

2. **Identify this study**
   Enter a useful study name and the sample, such as Zn-insulin. These labels do not infer or validate experimental settings.

3. **Create a new study**
   Click Create study... and choose a new empty folder. This creates the workspace; do not create it a second time in the study window.

4. **Or reopen a saved study**
   Click Open saved study... and choose the folder containing workspace.json. NanoScope restores its saved method and settings automatically.

Keep in mind

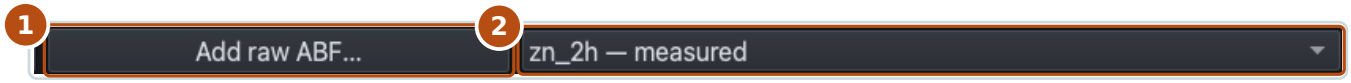
The profile supplies starting defaults. Apply every recording-specific recipe on the following pages. Saved workspaces retain their own numerical settings.

You should see

A named study opens with its staged controls. Continue selected study returns to an already open workspace.

Add recordings to your new study

Reference route · retain the automatic acceptance decisions



Reference capture, 21 Sep 2026. Workspace toolbar



Reference capture, 21 Sep 2026. Selected stage tab

DO THIS IN NANOSCOPE

1. **Add the recordings**
   Click Add raw ABF…. Select the four Zn ABFs listed on the Input files card, then confirm the file picker.

2. **Select the first recording**
   Use the recording selector at the top to choose the 2 h recording.

3. **Learn the two panels**
   The vertical tabs on the left run the stages. The right tabs show Trace / event, Results table, Saved figure and Methods / history. Scroll inside the LEFT panel to expose lower controls; drag the divider if needed.

Keep in mind

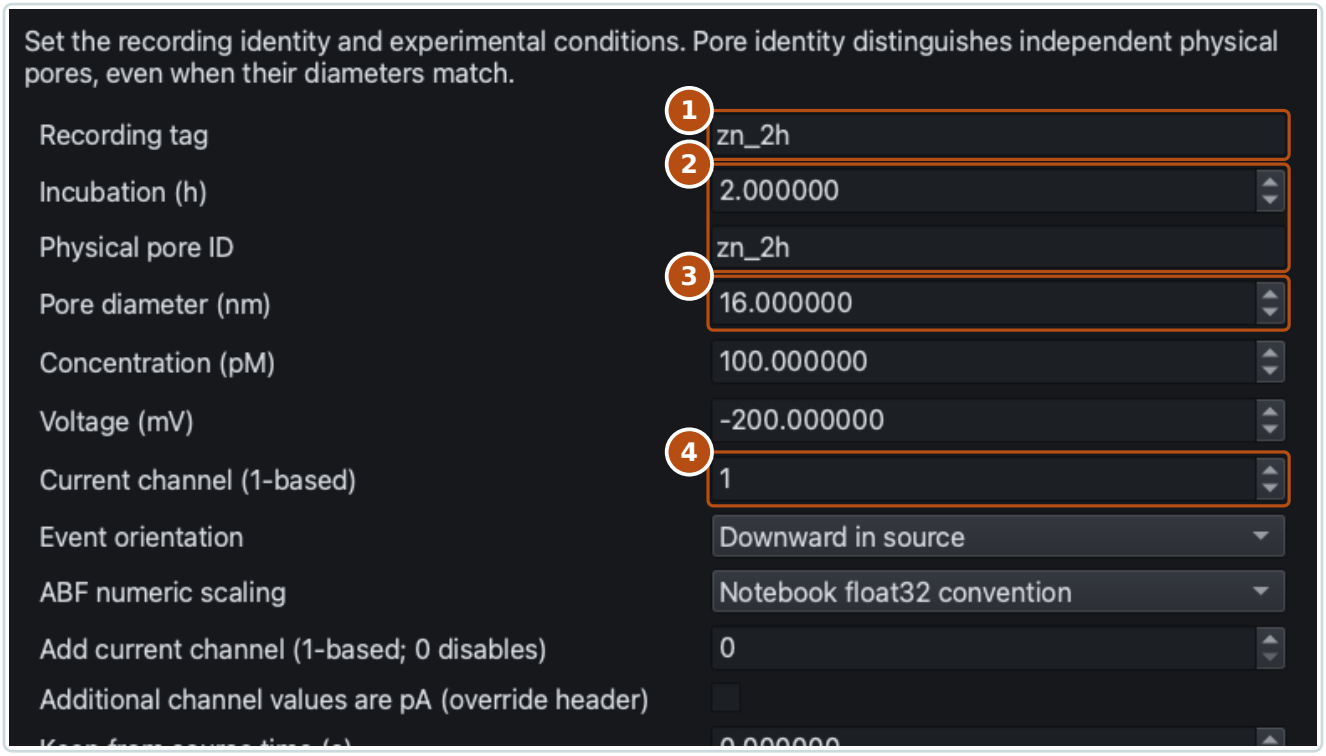
Create study... on the launcher has already created this folder. Add the exact reference ABFs listed in the input appendix. Choose a separate saved-workspace copy for alternative curation.

You should see

All four recordings appear in the selector. No study analysis has been run yet.

Identify the selected recording

Reference route · retain the automatic acceptance decisions



Reference capture, 21 Sep 2026. Controls · top of panel

Values for this step
2 h: tag / pore ID / incubation / diameter: zn_2h / zn_2h / 2 / 16 nm
10 h: zn_10h / zn_10h / 10 / 18 nm
18 h: zn_18h / zn_18h / 18 / 18 nm
30 h: zn_30h / zn_30h / 30 / 20 nm

DO THIS IN NANOSCOPE

1. **Check the recording tag**
   On 1 Sources, find Recording tag. Check that the source path below the form is the intended ABF.

2. **Enter the identity**
   For 2 h enter zn_2h for tag and pore ID, incubation 2. Use recipe cards R2h–R30h for the others.

3. **Enter the experiment**
   Set pore diameter 16 nm, concentration 100 pM and voltage -200 mV. Equal diameters can be different physical pores.

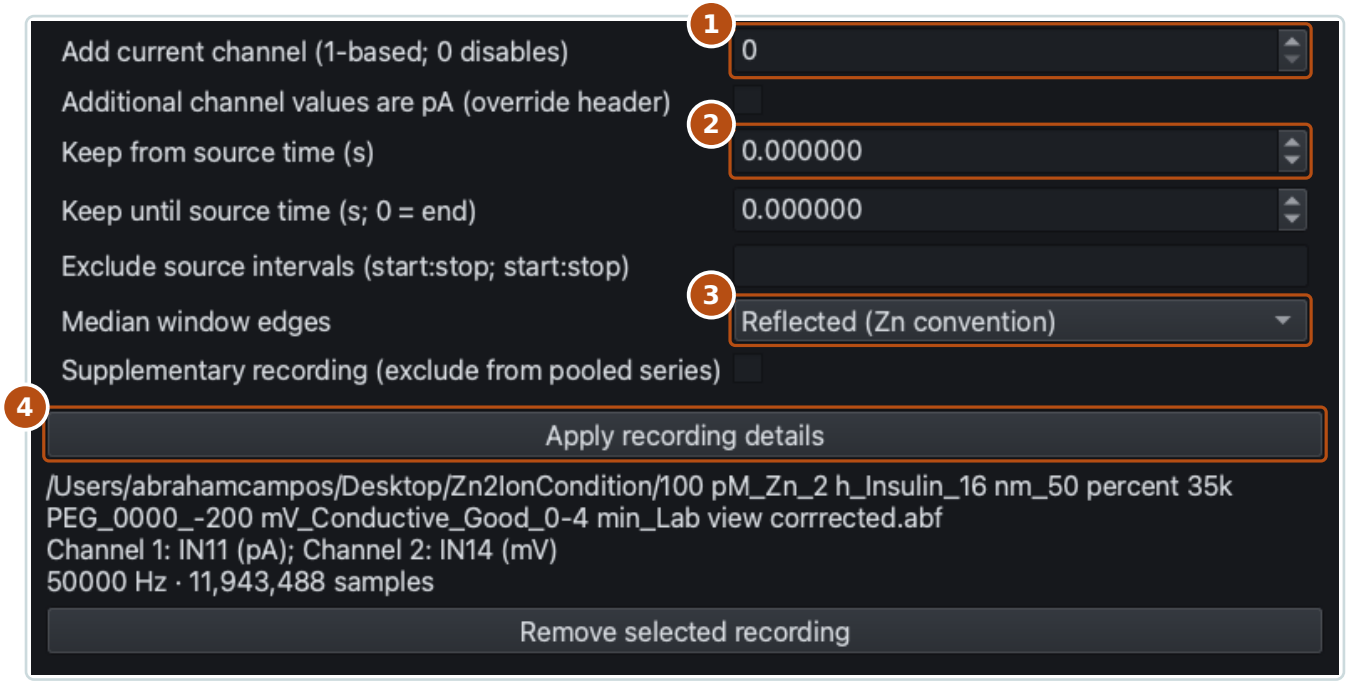
4. **Set channel and arithmetic**
   Choose current channel 1, Downward in source, and Notebook float32 convention. Continue to the lower controls on Z04 before applying.

Keep in mind
All files: 50,000 Hz. Channel 1 is IN11 in pA; channel 2 is IN14 in mV. Do not add it to the current.

You should see
Identity matches the selected ABF.

Keep the full recording and apply its details

Reference route · retain the automatic acceptance decisions



Reference capture, 21 Sep 2026. Controls · scroll down

Values for this step
Added channel / Additional pA override: 0 / Unchecked
Keep from / until / Exclude intervals: 0 / 0 (= end) / Blank
Median window edges: Reflected (Zn convention)
Supplementary recording: Unchecked

DO THIS IN NANOSCOPE

1. **Scroll the left panel down**
   Stay on 1 Sources. Find Add current channel and the time-range controls.

2. **Keep the Zn source settings**
   Set the controls exactly as shown in the card. Leave the exclusion field empty.

3. **Choose the median edges**
   Select Reflected (Zn convention). Leave Supplementary recording unchecked so this recording participates in the pooled series.

4. **Apply**
   Click Apply recording details. Check the sample count and duration against this recording’s recipe card.

Keep in mind

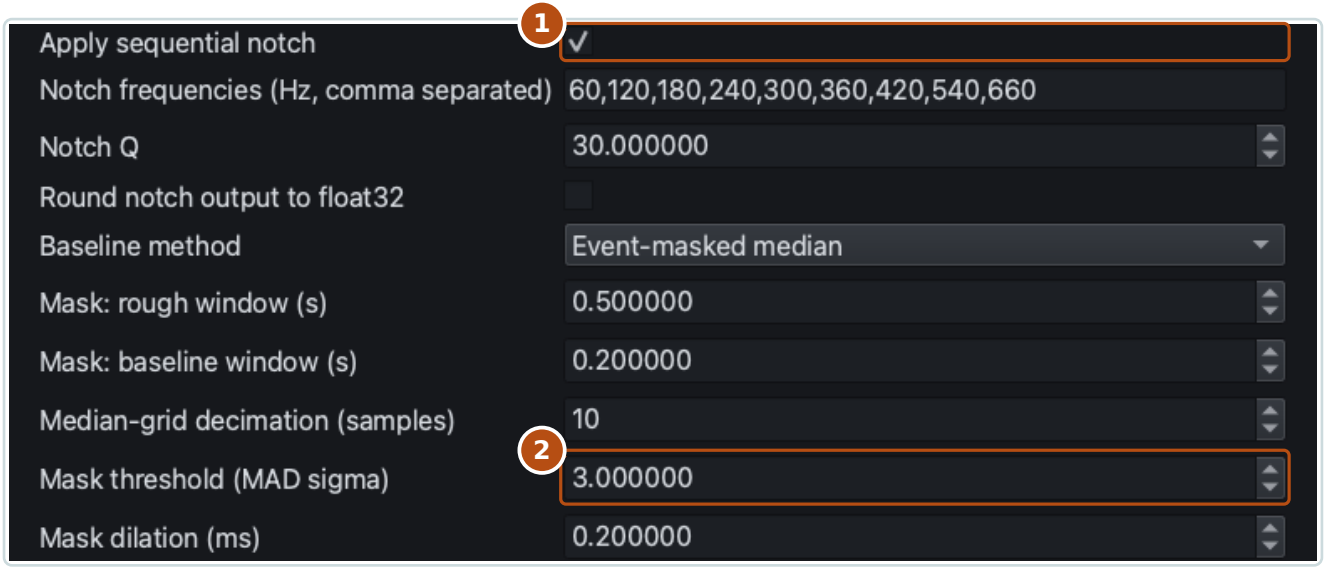
Notebook float32 convention reproduces the reader arithmetic; it is not an extra gain correction. Do not multiply by ten or invert based on the negative voltage.

You should see

The full recording is retained and its source identity is saved.

Prepare the raw current

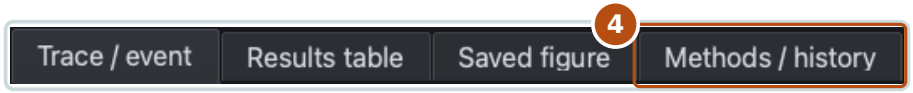
Reference route · retain the automatic acceptance decisions



Reference capture, 21 Sep 2026. Preparation controls · top



Reference capture, 21 Sep 2026. Scroll down · run preparation



Reference capture, 21 Sep 2026. Result tabs

Values for this step
Notch enabled / frequencies: Checked / 60,120,180,240,300,360,420,540,660 Hz
Notch Q / float32 rounding: 30 / Unchecked
Baseline / rough window / baseline window: Event-masked median / 0.5 s / 0.2 s
Median-grid decimation / Despike: 10 samples / 0

DO THIS IN NANOSCOPE

1. **Enable the Zn notch**
   Click 2 Prepare and check Apply sequential notch. Enter the common settings in this card.

2. **Set the per-recording mask**
   For zn_2h use threshold 3.0 and dilation 0.2 ms. For zn_10h use 3.5 and 2.0 ms; for zn_18h and zn_30h use 3.0 and 2.0 ms.

3. **Run preparation**
   Scroll the left panel down and click Prepare selected recording. Wait until the operation finishes before changing stages.

4. **Check the saved method**
   Open Methods / history on the right and confirm source, settings and preparation checks.

Keep in mind

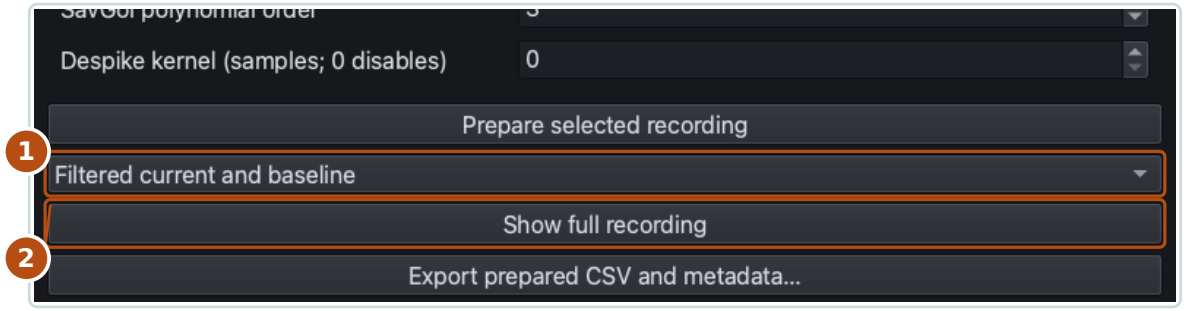
Other baseline-method fields do not replace the mask windows. No extra low-pass, high-pass, moving average or despiking is part of this reference.

You should see

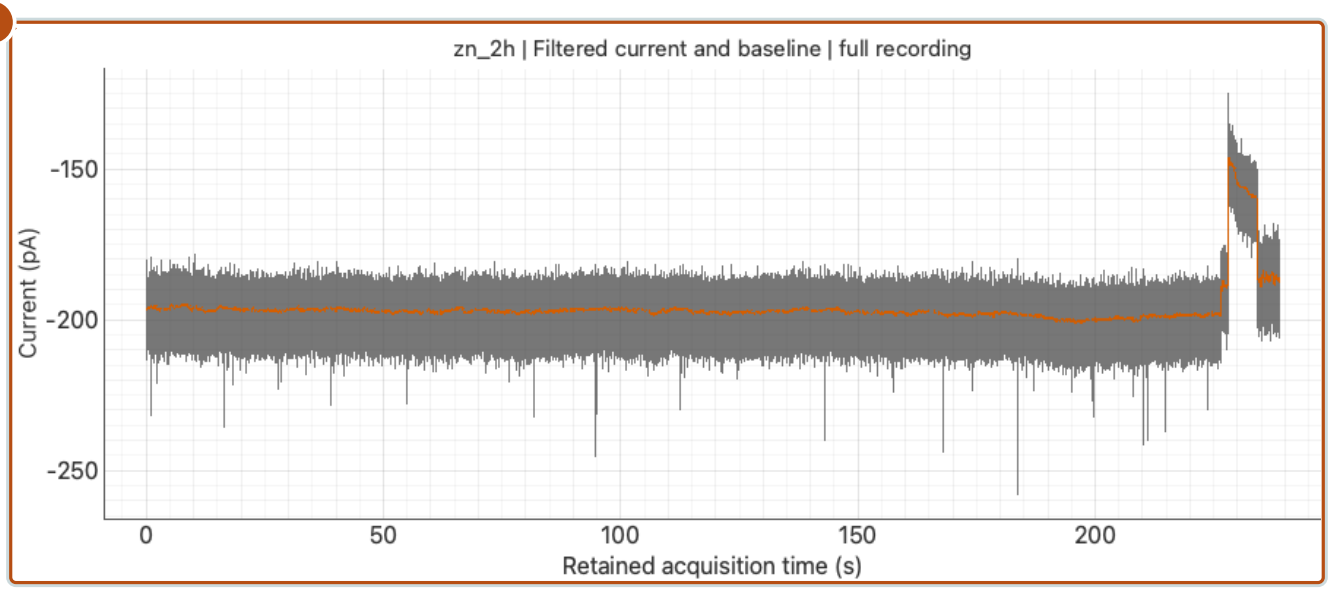
Prepared current is baseline-subtracted current in pA. Model normalization has not happened yet.

Inspect raw, baseline and prepared traces

Reference route · retain the automatic acceptance decisions



Reference capture, 21 Sep 2026. Controls · scroll down



Reference capture, 21 Sep 2026. Saved reference result

DO THIS IN NANOSCOPE

1. **Show the raw current**
   On 2 Prepare, scroll down to the preview selector. Choose Raw current, then click Show full recording.

2. **Inspect the baseline**
   Choose Filtered current and baseline, then click Show full recording. Zoom into quiet sections and events; the baseline should follow slow drift without diving into brief events.

3. **Inspect the model input**
   Choose Prepared current. Inspect both short events and quiet sections in Trace / event.

Keep in mind

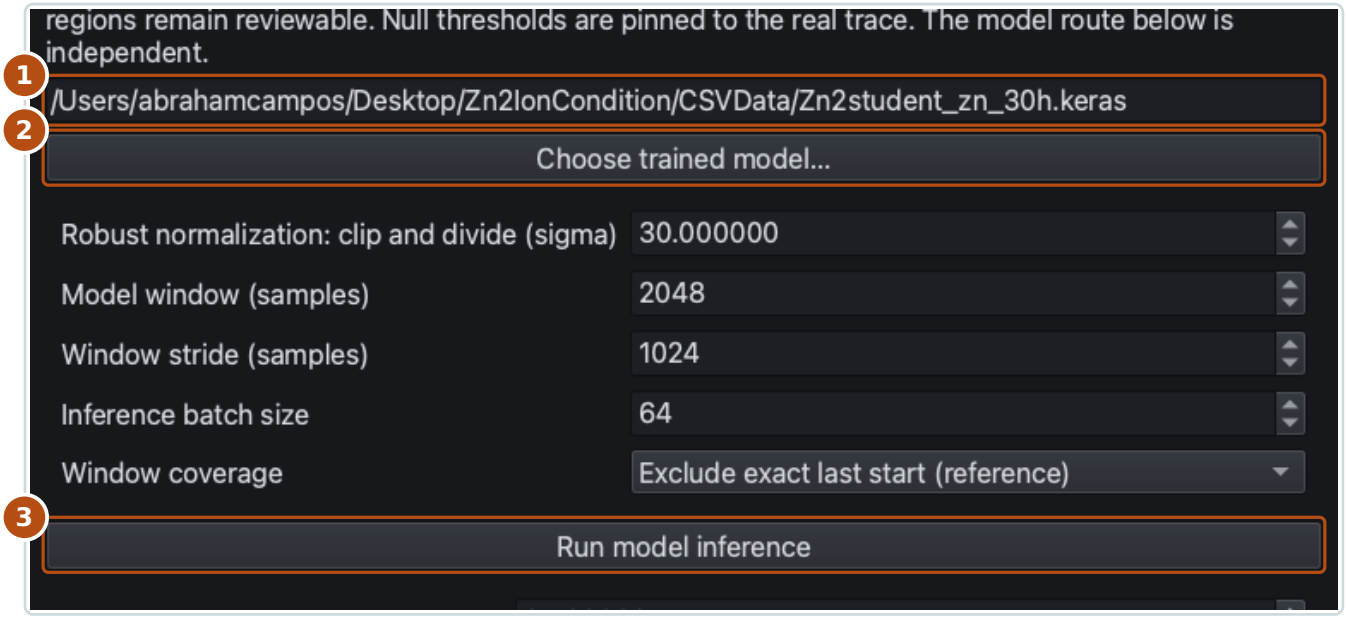
The overview preserves extrema. Use Methods / history for recorded quality checks; do not tune those checks to force a desired event count. Optional Export prepared CSV and metadata… writes <tag>.csv.gz and <tag>_metadata.json; no export is required to continue.

You should see

You can distinguish the raw open-pore level, notched current, estimated baseline and prepared signal.

Run the supplied student model

Reference route · retain the automatic acceptance decisions



Reference capture, 21 Sep 2026. Controls · scroll down

Values for this step
Robust normalization: clip and divide (sigma): 30
Model window / Window stride (samples): 2048 / 1024
Inference batch size: 64
Window coverage: Exclude exact last start (reference)

DO THIS IN NANOSCOPE

1. **Find the trained-model controls**
   Click 3 Detect / review. Scroll DOWN past event review, signal-detector settings and null controls to Choose trained model…. The signal controls belong to another route.

2. **Choose the same model for every file**
   Click Choose trained model… and select CSVData/Zn2student_zn_30h.keras from the input folder. Enter the inference settings below.

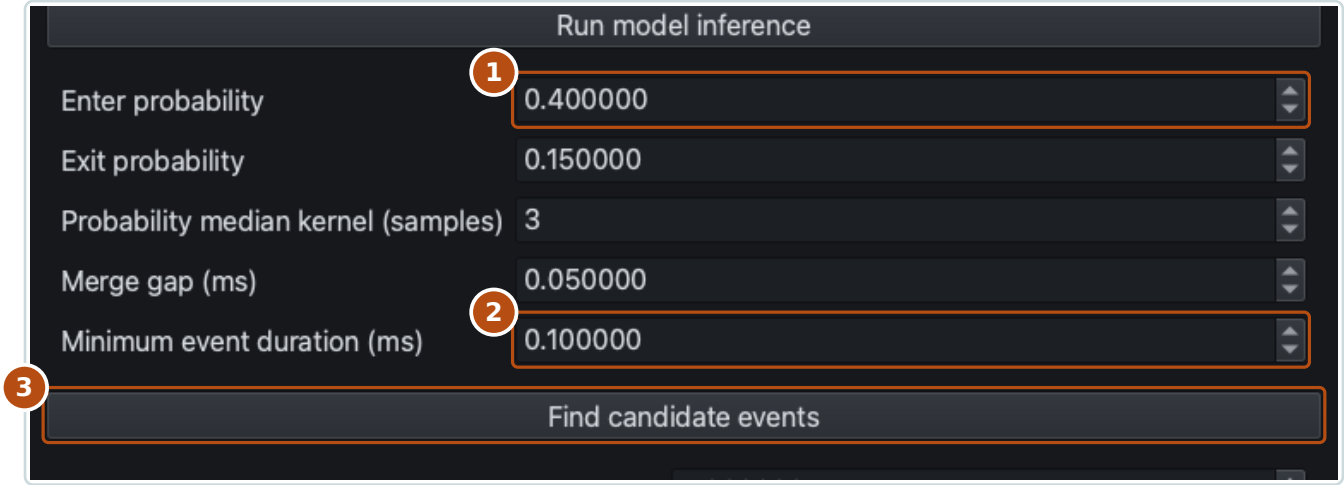
3. **Run inference**
   Click Run model inference and wait. Confirm the model path when switching recordings.

Keep in mind
The 30 h model is used for every recording. Inspect probabilities through 2 Prepare → Model probability → Show full recording.

You should see
Probabilities are saved. The reference leaves 576 samples uncovered at 2 h and 1,023 at the other times; those tails are unmeasured.

Convert probabilities to candidate events

Reference route · retain the automatic acceptance decisions



Reference capture, 21 Sep 2026. Controls · scroll down

Values for this step
Enter / Exit probability: 0.4 / 0.15
Probability median kernel: 3 samples
Merge gap / Minimum duration: 0.05 ms / 0.1 ms

DO THIS IN NANOSCOPE

1. **Find the candidate controls**
   Stay on 3 Detect / review and scroll below Run model inference to Enter probability.

2. **Enter the five values**
   Use enter 0.4, exit 0.15, median kernel 3 samples, merge gap 0.05 ms and minimum duration 0.1 ms.

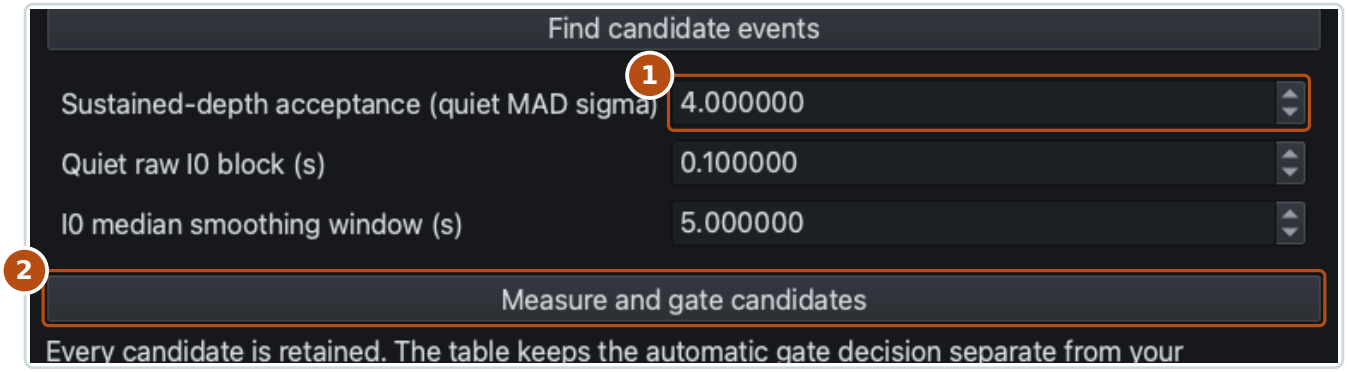
3. **Find candidates**
   Click Find candidate events and wait. This reads saved probabilities without running the model again.

Keep in mind
At 50,000 Hz, the minimum duration is five samples. The reference does not snap onsets or trim offsets. Nearby candidates merge before the duration test.

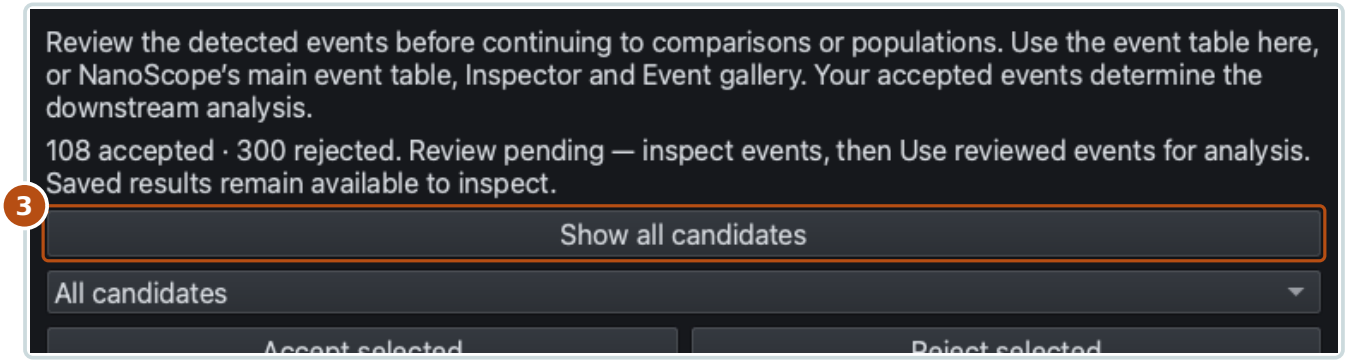
You should see
Candidate intervals have been found, but amplitude acceptance still needs to be measured.

Measure depth and apply the automatic gate

Reference route · retain the automatic acceptance decisions



Reference capture, 21 Sep 2026. Controls · scroll down



Reference capture, 21 Sep 2026. Controls · top of panel

Values for this step
Sustained-depth acceptance (quiet MAD sigma): 4
Quiet raw I0 block: 0.1 s
I0 median smoothing window: 5 s

DO THIS IN NANOSCOPE

1. **Scroll to measurement**
   Below Find candidate events, locate Sustained-depth acceptance and the two raw-I0 controls.

2. **Set and measure**
   Enter the values below, then click Measure and gate candidates. Wait for completion.

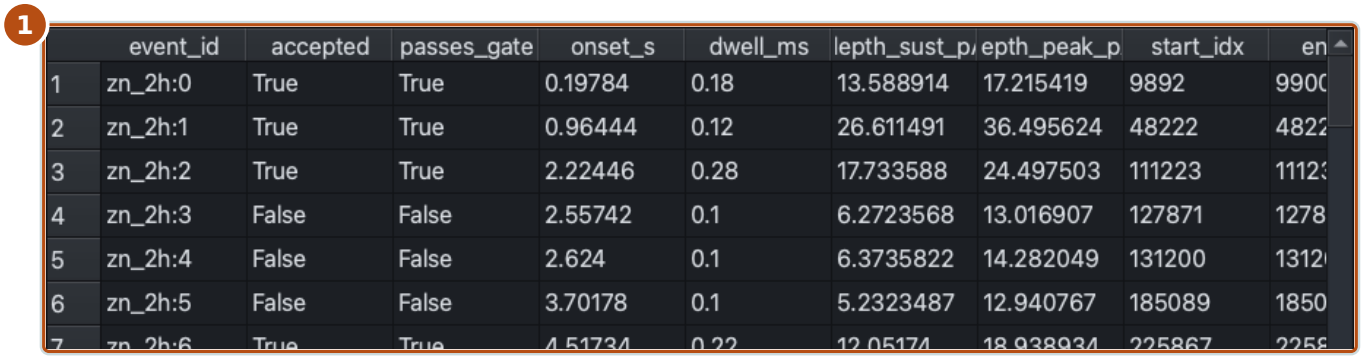
3. **Return to the event list**
   Scroll the left panel to the TOP and click Show all candidates.

Keep in mind
passes_gate records the automatic sustained-depth decision. accepted initially copies that decision and controls downstream analysis. All candidates, including rejects, remain in the table.

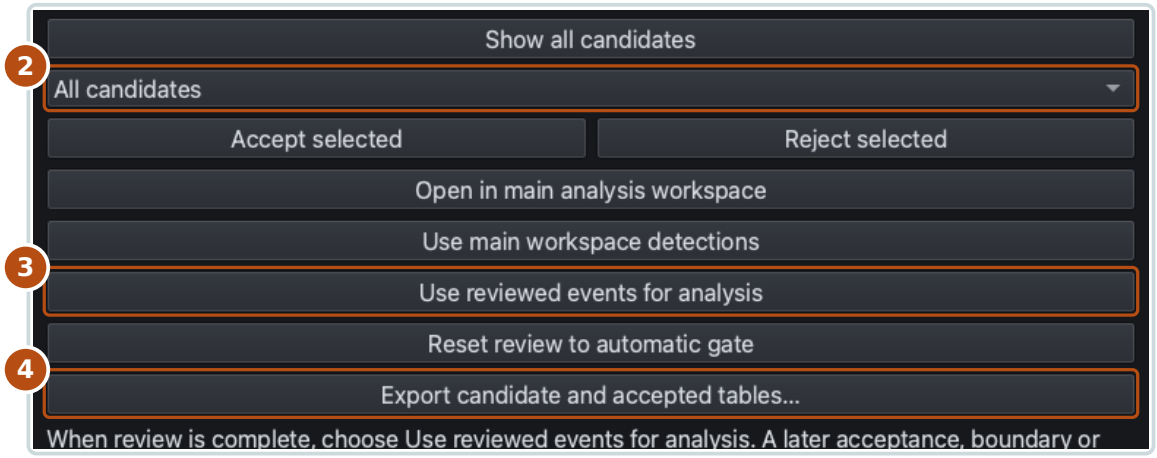
You should see
For zn_2h: 408 candidates and 108 accepted events. Use the recipe card for each other recording.

Inspect events and confirm the reference selection

Reference route · retain the automatic acceptance decisions



Reference capture, 21 Sep 2026. Saved reference result



Reference capture, 21 Sep 2026. Controls · scroll down

DO THIS IN NANOSCOPE

1. **Choose an event**
   With Show all candidates open, click a Results table row. Its waveform appears in Trace / event. Scroll the table horizontally to see bounds, measurements and accepted.

2. **Inspect both decisions**
   Use the filter to inspect Accepted only and Rejected only. For the reference route, leave acceptance unchanged.

3. **Confirm the selected events**
   Click Use reviewed events for analysis. This confirms the current selection; it does not claim every event was individually inspected.

4. **Save a table if needed**
   Export candidate and accepted tables… writes both <tag>_candidates.csv and <tag>_accepted.csv.

Keep in mind

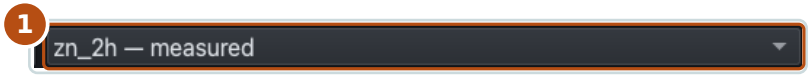
For a curated alternative, select rows and use Accept selected / Reject selected; see the optional curation pages. Shift selects a range and Command selects separate rows. Reset review to automatic gate discards manual acceptance and current human-review status.

You should see

Review is confirmed for this recording. The shaded interval is the measured event; the dot marks its actual minimum sample.

Repeat the per-recording loop four times

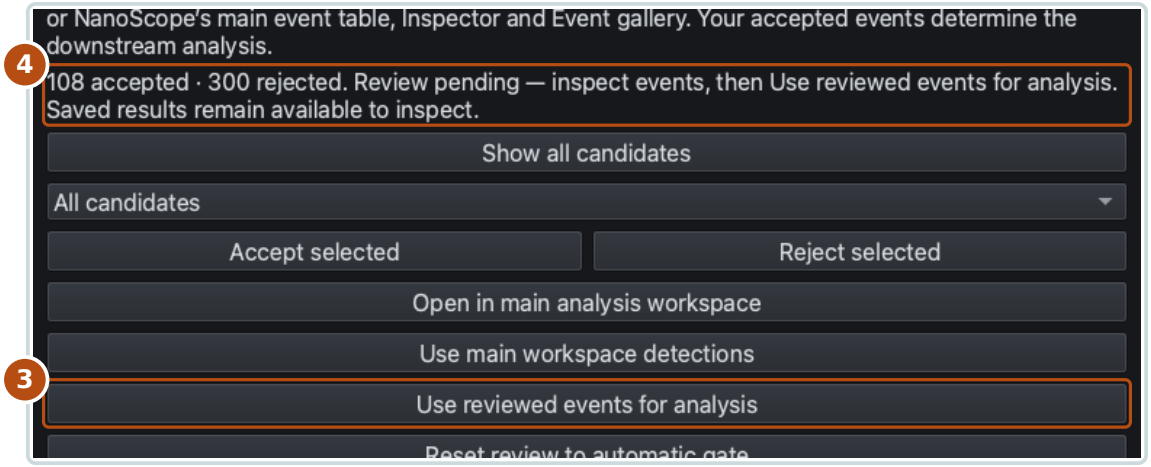
Reference route · retain the automatic acceptance decisions



Reference capture, 21 Sep 2026. Workspace toolbar



Reference capture, 21 Sep 2026. Selected stage tab



Reference capture, 21 Sep 2026. Controls · top of panel

Values for this step
zn_2h: candidates / accepted: 408 / 108
zn_10h: 3,943 / 110
zn_18h: 1,647 / 194
zn_30h: 11,004 / 1,076

DO THIS IN NANOSCOPE

1. **Select the next file**
   Use the recording selector to choose zn_10h, then zn_18h, then zn_30h.

2. **Repeat Sources through measurement**
   Repeat Z03–Z09 for that recording, using its recipe card. Check source settings and model path rather than assuming they carried over.

3. **Inspect and confirm**
   Repeat Z10: inspect events, retain automatic acceptance and click Use reviewed events for analysis for EACH file.

4. **Check all four totals**
   Compare your counts with this card. If a count differs unexpectedly, check inputs and saved stage settings before continuing.

Keep in mind

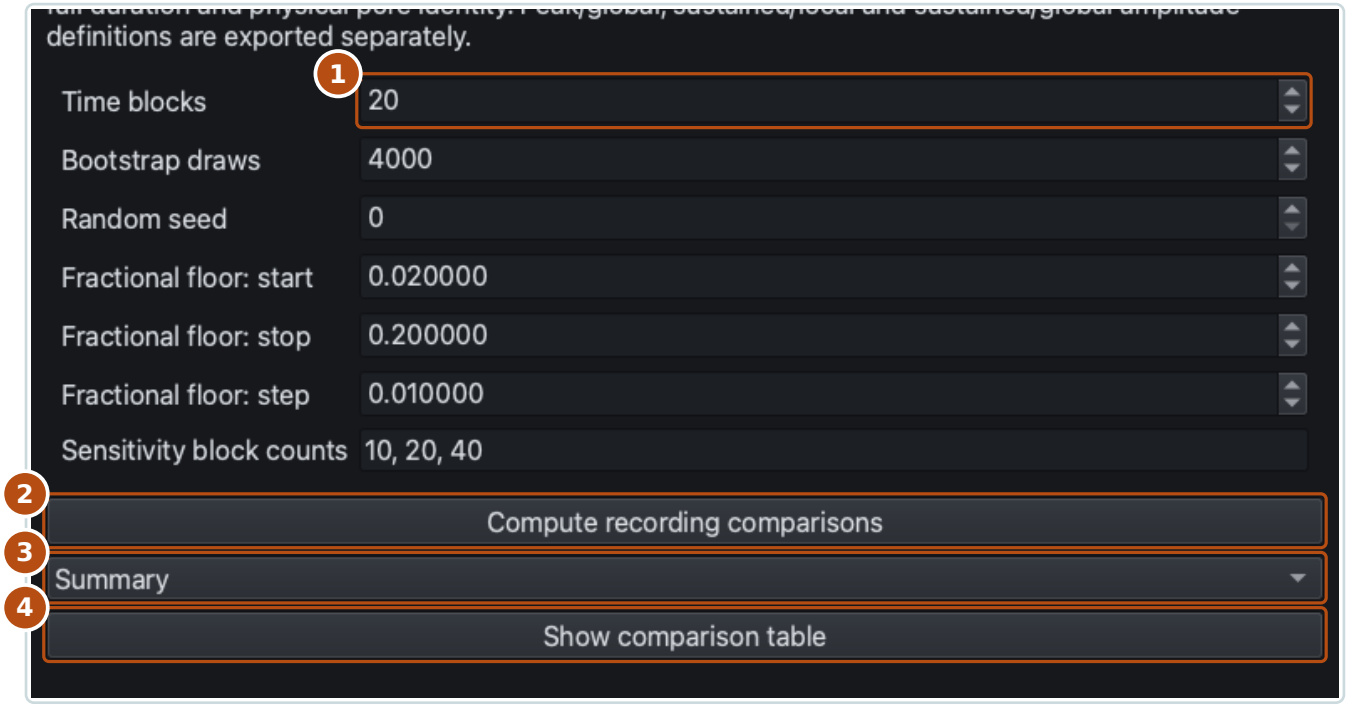
Total: 17,002 candidates / 1,488 accepted. Historical 10 h CSV has two extra rejected candidates (A05). Restore completed settings reveals saved parameters; editing fields alone does not recompute.

You should see

Every recording is measured and review-confirmed. Follow Z12, or use the optional curated branch.

Compute recording comparisons

Reference route · retain the automatic acceptance decisions



Reference capture, 21 Sep 2026. Controls · top of panel

Values for this step
Time blocks / Bootstrap draws / Random seed: 20 / 4000 / 0
Fractional floor start / stop / step: 0.02 / 0.20 / 0.01
Sensitivity block counts: 10,20,40

DO THIS IN NANOSCOPE

1. **Set the comparison blocks**
   On 4 Compare, set Time blocks to 20 after confirming every included recording. For a curated branch, finalize acceptance first.

2. **Enter comparison settings**
   Use the values in this card; then click Compute recording comparisons and wait.

3. **Inspect Summary**
   Choose Summary in the comparison-table selector and click Show comparison table. Check counts, durations, pore IDs and rates.

4. **Inspect the sensitivity tables**
   Choose rate_block_sensitivity, gate_transfer_sweep, gate_transfer_pairs and distribution_distances in turn. The saved table changes without recalculating.

Keep in mind

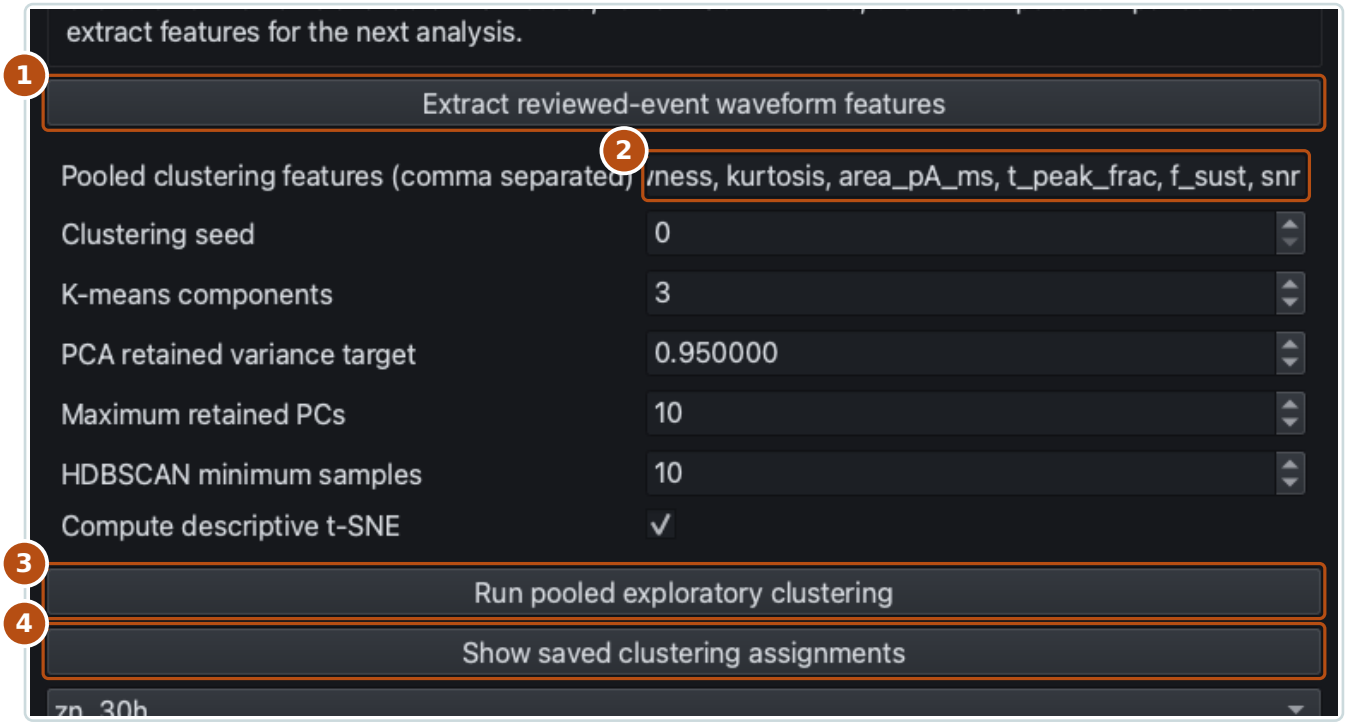
Rate = accepted count / duration. Time-block intervals describe within-recording uncertainty, not independent pore replication. Use A03 for numeric checkpoints and A04 for amplitude meanings.

You should see

Rates are approximately 0.452129, 0.538186, 1.637452 and 4.774109 events/s (2, 10, 18 and 30 h).

Extract features and run pooled clustering

Reference route · retain the automatic acceptance decisions



Reference capture, 21 Sep 2026. Controls · scroll down

Values for this step
Clustering seed / K-means components: 0 / 3
PCA retained variance / Maximum retained PCs: 0.95 / 10
HDBSCAN minimum samples: 10
Compute descriptive t-SNE: Checked

DO THIS IN NANOSCOPE

1. **Find feature extraction**
   Click 5 Features / populations. Scroll DOWN past Review whole clusters to Extract reviewed-event waveform features. Click it and wait.

2. **Enter the feature list**
   Paste the 12-column Pooled feature list from A02 into Pooled clustering features (comma separated). Use the remaining values below.

3. **Run clustering**
   Click Run pooled exploratory clustering and wait.

4. **Inspect assignments**
   Click Show saved clustering assignments. Inspect PCA, K-means and HDBSCAN labels; t-SNE is descriptive.

Keep in mind

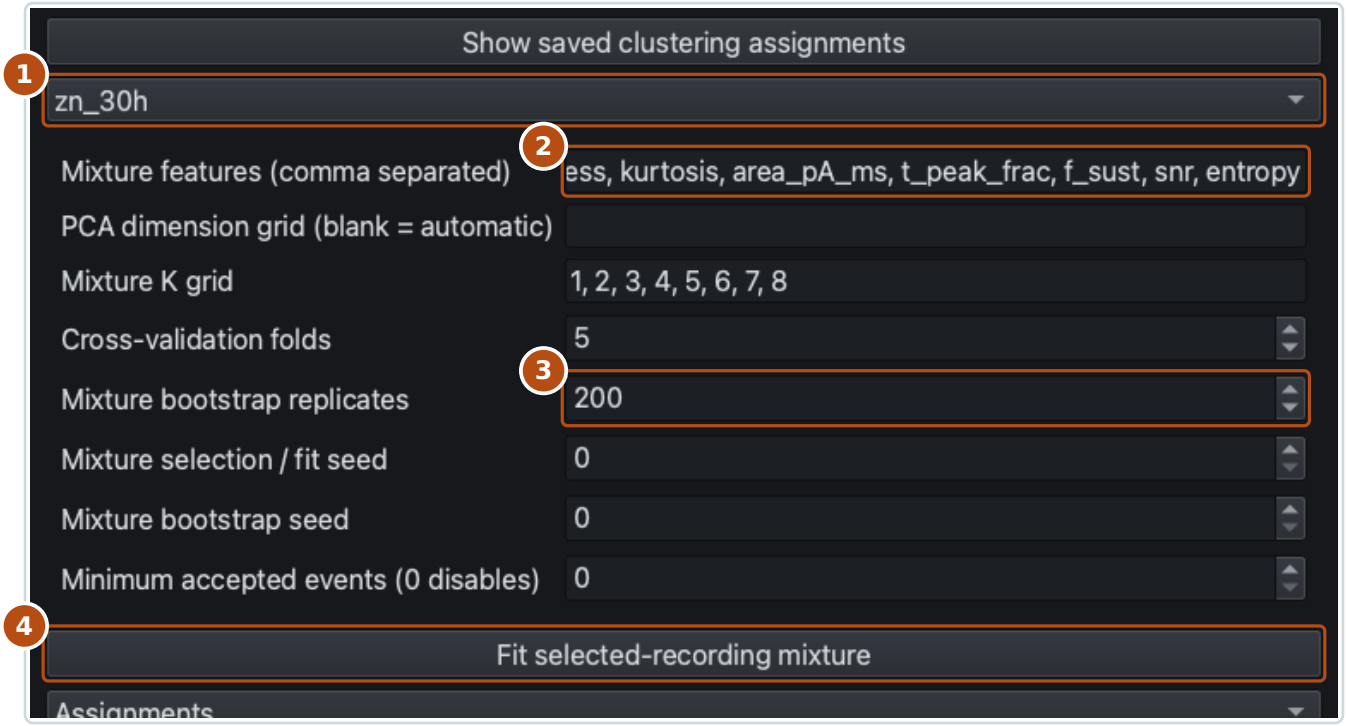
Reference feature extraction yields 1,488 events. HDBSCAN Noise (-1) is a label, not an automatic rejection. For whole-cluster curation, use C04–C05 now; afterward re-extract features and rerun the final analyses.

You should see

A pooled exploratory result exists. Its groups are review aids, not established molecular identities.

Fit the separate 30 h mixture

Reference route · retain the automatic acceptance decisions



Reference capture, 21 Sep 2026. Controls · scroll down

Values for this step
PCA dimension grid: 2,3,4,5,6,7,8,9,10,11,12,13
Mixture K grid: 1,2,3,4,5,6,7,8
Cross-validation folds / Bootstrap replicates: 5 / 200
Selection-fit seed / Bootstrap seed / Minimum events: 0 / 0 / 0

DO THIS IN NANOSCOPE

1. **Find the mixture controls**
   On 5 Features / populations, scroll DOWN past pooled clustering and select zn_30h in the mixture-target selector.

2. **Use the 13-feature mixture list**
   Paste the Mixture feature list from A02, including entropy at the end. Enter the explicit grids below.

3. **Set the fit and bootstrap**
   Set folds 5, bootstrap replicates 200, both seeds 0 and minimum accepted events 0.

4. **Fit**
   Click Fit selected-recording mixture and wait. The reference fits the 1,076 accepted zn_30h events.

Keep in mind

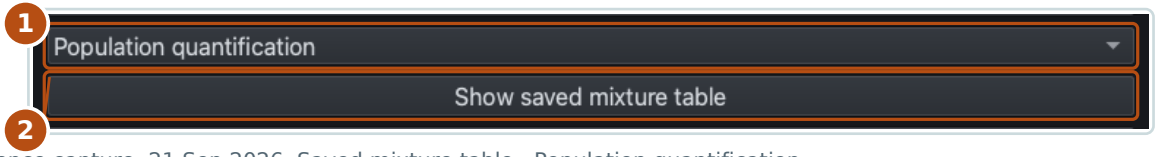
The explicit PCA grid equals the reference's automatic 2–13 grid. Minimum events 0 disables only the user threshold; fitting still requires at least 60 feature events.

You should see

Saved results include model selection, assignments, weights, bootstrap and diagnostics.

Inspect the mixture before making figures

Reference route · retain the automatic acceptance decisions



Reference capture, 21 Sep 2026. Saved mixture table · Population quantification

3

	subpop	n	assigned_event_fraction	mixture_weight_lo95	mixture_weight_hi95	assigned_event_rate_per_s	d
1	0	387	0.35966543	0.18248682	0.63811254	1.7170819	0.
2	1	295	0.27416357	0.15198012	0.65987017	1.3088867	0.
3	2	269	0.25	0.11112949	0.26796768	1.1935272	2.
4	3	125	0.116171	0.01394052	0.11809083	0.55461301	1.

Reference capture, 21 Sep 2026. Saved 30 h population quantification · four components

Values for this step
Verified reference selection: 11 PCs; 4 components
Reference hard assignment counts: 387, 295, 269, 125
Saved labels / trace display labels: subpop 0–3 / P1–P4

DO THIS IN NANOSCOPE

1. **Read model selection**
   Scroll below Fit selected-recording mixture. Choose Cross-validation grid, then Show saved mixture table.

2. **Read populations**
   Choose Population quantification. Distinguish assigned event fraction/rate from the fitted soft mixture weight.

3. **Read assignments and bootstrap**
   Choose Assignments and Mixture-weight bootstrap in turn. Retain any weak-separation or label-switching warnings.

Keep in mind

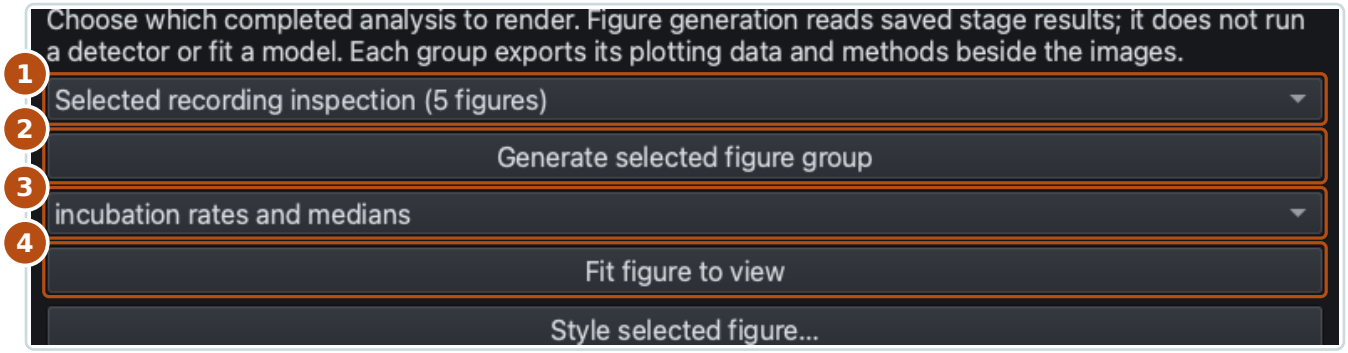
Treat the checks as observations, not parameters to force. Equivalent fits can permute component numbers. The reference is weakly separated (silhouette ≈0.11); four fitted components do not establish four molecular species. Bootstrap intervals describe mixture weights, not hard-count uncertainty.

You should see

Diagnostics and event geometry support the saved result; you have not accepted a count match alone.

Generate the complete 32-figure set

Reference route · retain the automatic acceptance decisions



Reference capture, 21 Sep 2026. Controls · top of panel

Values for this step
Recording inspection: 4 × 5 = 20 figures
Comparisons: 7 figures
Pooled clustering, with t-SNE: 2 figures
30 h mixture: 3 figures; reference total 32

DO THIS IN NANOSCOPE

1. **Generate inspection figures**
   Click 6 Figures. Select Selected recording inspection (5 figures), then Generate selected figure group. Repeat for each recording using the top selector.

2. **Generate the other groups**
   Generate Recording comparisons, Pooled exploratory clustering and Selected-recording mixture once each.

3. **Select a saved figure**
   Choose it in the saved-figure selector. Names, not report numbers, identify the figures.

4. **Inspect at a readable scale**
   Click Fit figure to view, then zoom into dense or tall galleries.

Keep in mind

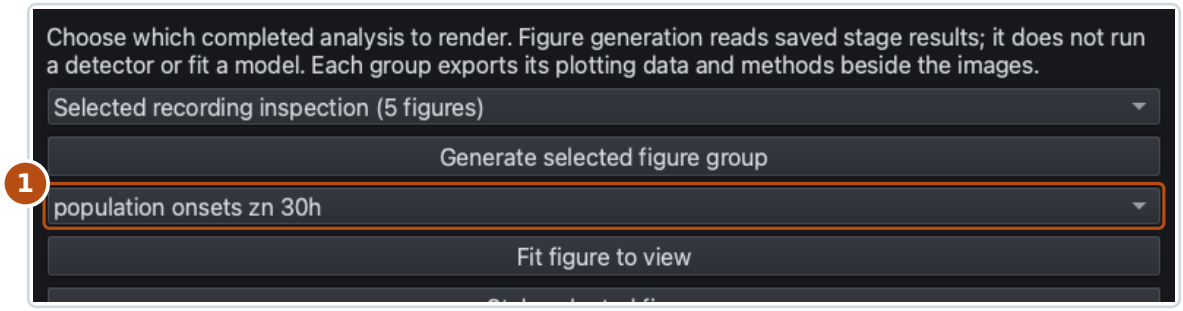
Finish all scientific operations before this step. New upstream results clear dependent current figures. Generation reads completed stages; it does not rerun inference or refit. Figure files and numerical sidecars are in timestamped runs folders.

You should see

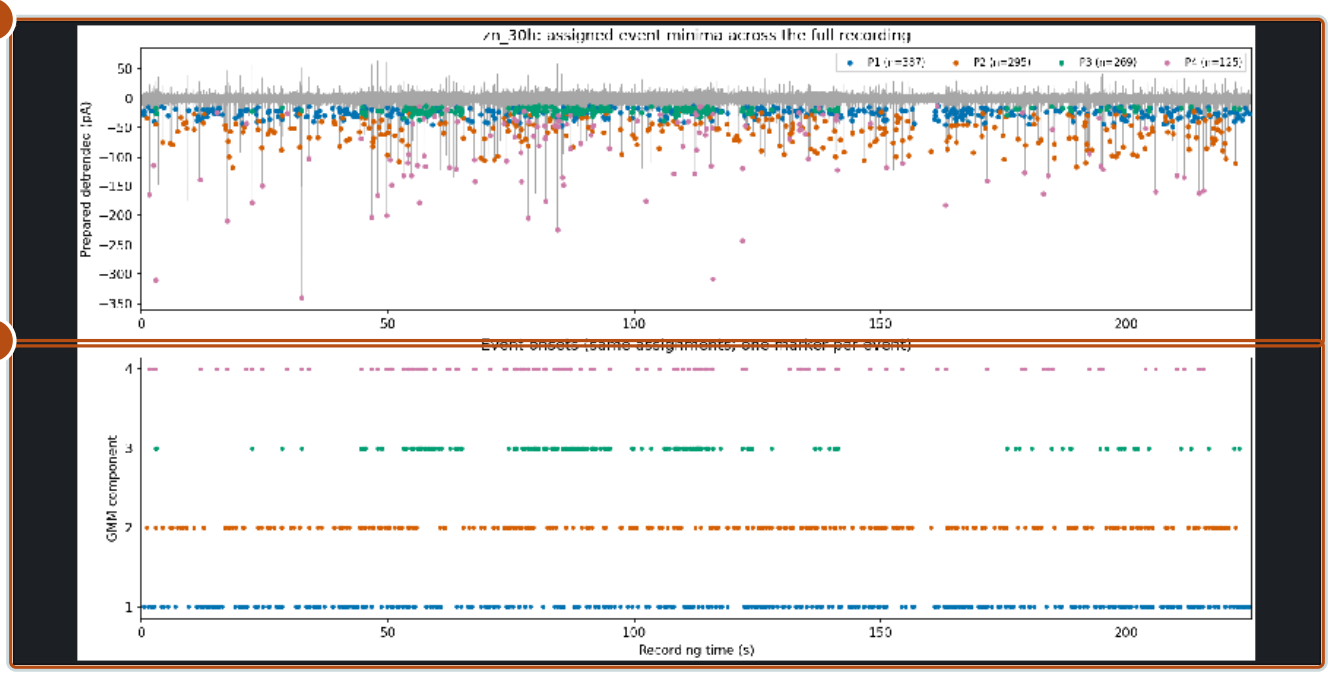
All 32 named figures are available. The inventory card identifies each type.

Check the assignment trace against event bounds

Reference route · retain the automatic acceptance decisions



Reference capture, 21 Sep 2026. Controls · top of panel



Reference capture, 21 Sep 2026. Saved reference result

DO THIS IN NANOSCOPE

1. **Open the assignment figure**
   On 6 Figures, select population_onsets_zn_30h. This is the former report's Figure 32.

2. **Inspect upper markers**
   Zoom into a few upper-trace markers. Each dot is the actual current minimum INSIDE its assigned event, on the recording's clock.

3. **Compare onset markers**
   The lower timeline marks event START times. A start marker and a minimum marker need not have identical time positions.

Keep in mind

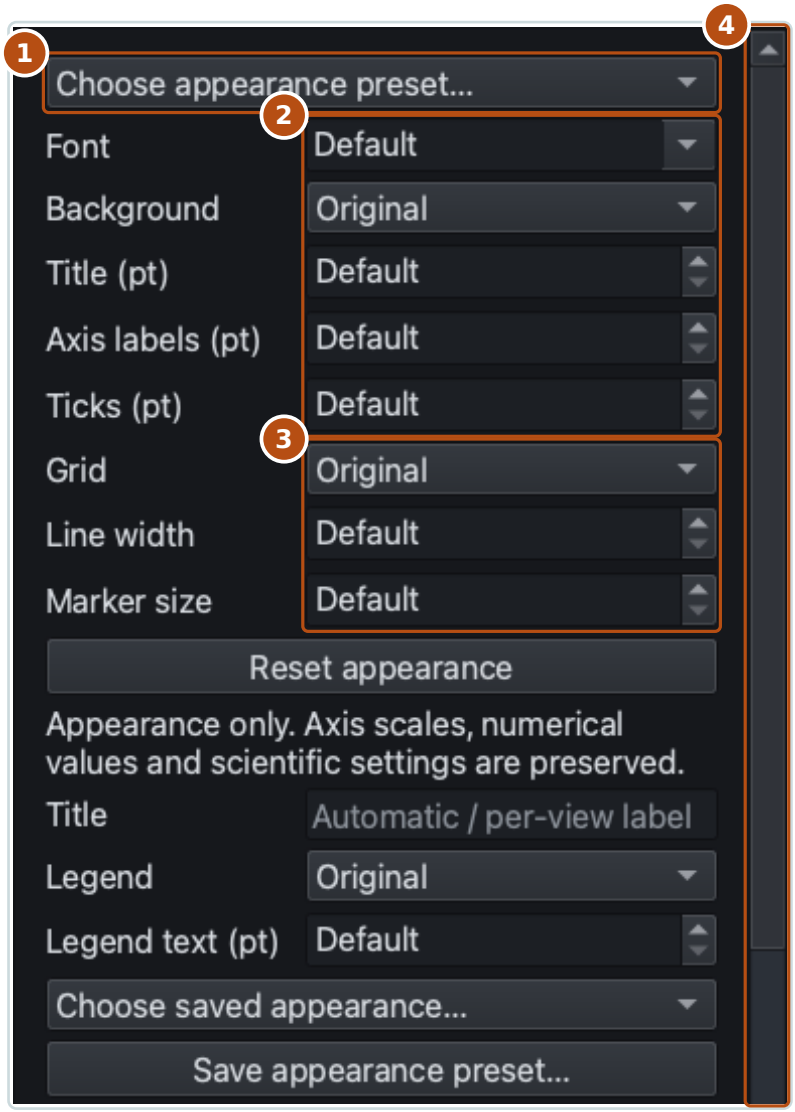
To cross-check: 5 Features / populations → Saved mixture table: Assignments → Show saved mixture table. Match tag and sample bounds; inspect the event on 3 Detect / review. The _events.csv plotting sidecar records marker coordinates.

You should see

Assigned samples, event bounds and trace geometry agree; markers are not interpreted as arbitrary baseline points.

Choose readable fonts and appearance

Reference route · retain the automatic acceptance decisions



Reference capture, 21 Sep 2026. NanoScope reference controls and saved results.

DO THIS IN NANOSCOPE

1 Open the editor
From 6 Figures select a saved figure and click Style selected figure…. In the editor shown here, choose Publication · light or your preferred preset.

2 Set text sizes
Set Font, Title (pt), Axis labels (pt) and Ticks (pt). Choose sizes readable at the final publication size.

3 Set plot appearance
Adjust Background, Grid, Line width, Marker size and legend controls. These are appearance settings; event choices and fitted populations remain unchanged.

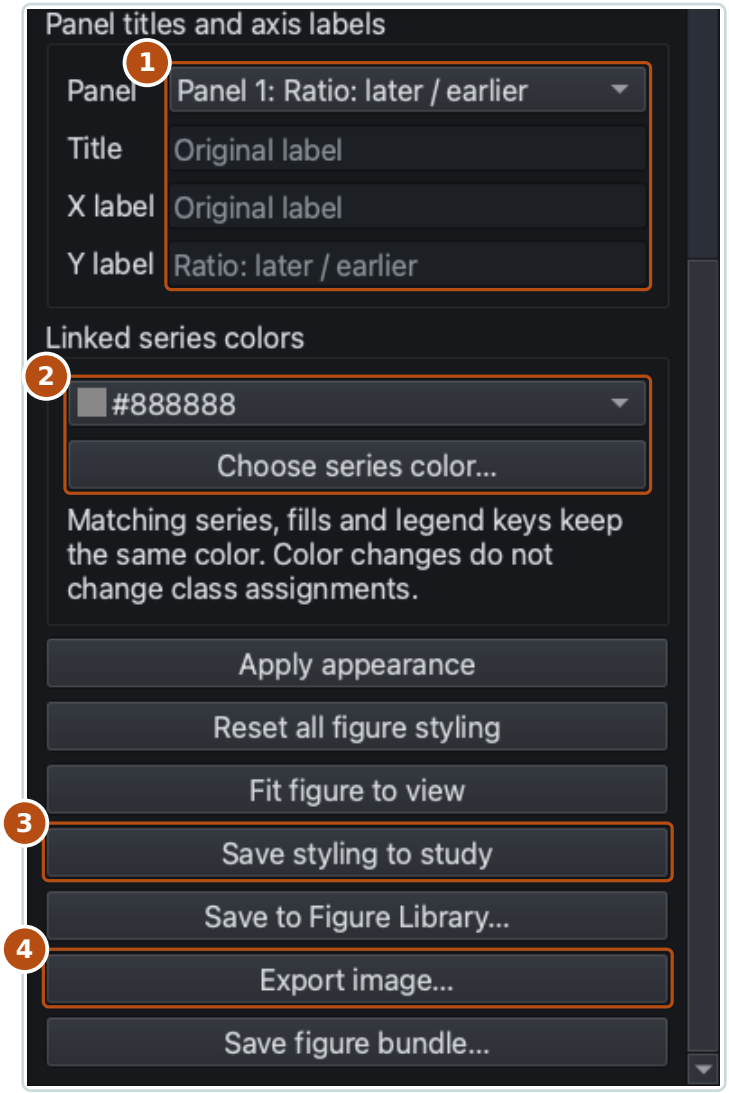
4 Continue lower down
Scroll DOWN inside the editor’s left panel for Panel titles and axis labels, Linked series colors and the Apply/Save buttons. Continue on Z18b.

Keep in mind
The editor does not expose arbitrary scientific axis ranges, axis scales or plot-type changes.

You should see
The desired appearance is selected; commit it on the next page.

Save labels, colors and final figure files

Reference route · retain the automatic acceptance decisions



Reference capture, 21 Sep 2026. NanoScope reference controls and saved results.

DO THIS IN NANOSCOPE

1. **Edit panel labels**
   In Panel titles and axis labels, choose a panel and edit its title or labels. Preserve the units and scientific meaning.

2. **Choose linked colors**
   Select a series in Linked series colors and click Choose series color…. Matching series, fills and legend keys change together.

3. **Preview, then commit**
   Click Apply appearance and inspect the result. Click Save styling to study so the style returns when reopening or regenerating this figure.

4. **Export the figure**
   Click Export image… for an image or Save figure bundle… for the figure, plotting data and provenance. Repeat for your selected figures.

Keep in mind

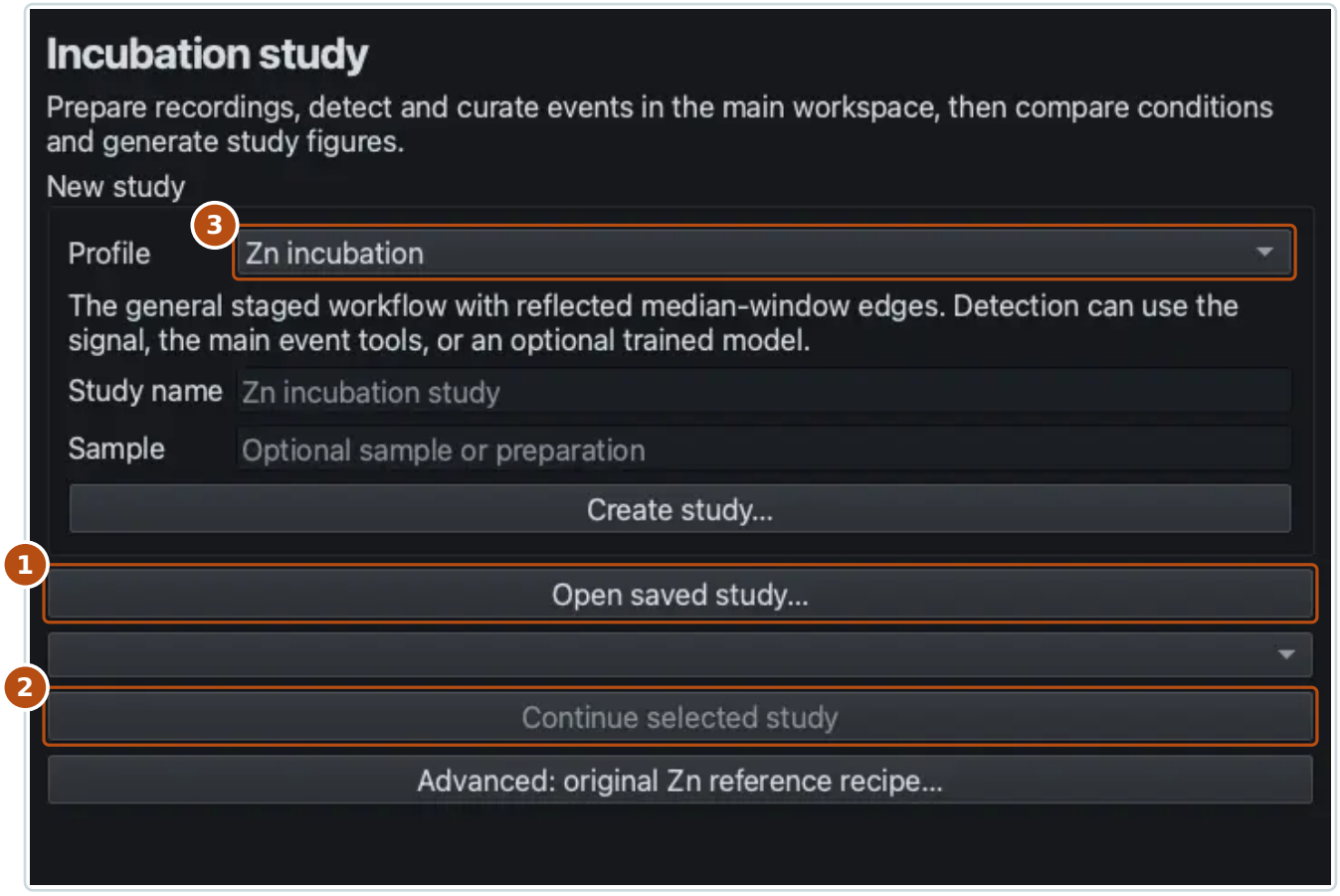
Apply appearance alone only previews. For optional multi-panel composition use Save to Figure Library… and follow A08; save the main project separately. Recheck colors after refitting changes cluster labels.

You should see

Appearance is committed to the study and the selected export files are saved.

Save and reopen the complete study

Unified incubation workflow | NanoScope 40.15.59



Current unified study launcher · zn

DO THIS IN NANOSCOPE

1. **Reopen the saved folder**
   In the study, finish any operation and click Save workspace. Close it, then open Analysis > Incubation study... > Open saved study.... Select the whole workspace folder.

2. **Continue an open study**
   If the study is already open, select it in Open study workspaces and click Continue selected study.

3. **Keep the saved method**
   The Profile control is for new studies. Opening a saved study selects the proper scientific method without applying a different profile.

Keep in mind

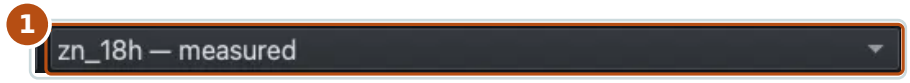
Keep workspace.json, arrays, runs and supporting files together. Preserve original ABFs and models for reruns. Use one editing session per folder. Save the main project separately for main curation and Figure Library entries.

You should see

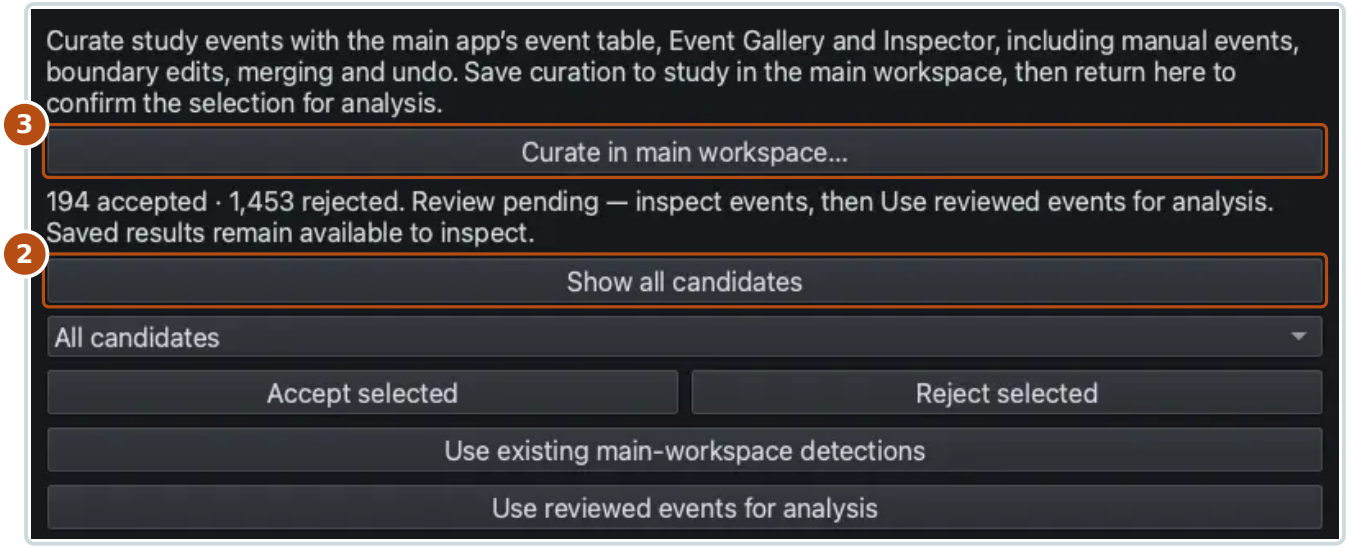
Check restored event decisions, committed settings, comparison tables, fits and styled figures without rerunning completed stages.

Curate with the main event tools

Unified incubation workflow | NanoScope 40.15.59



Current recording selector, cropped from the study toolbar.



Current build 40.15.59. ZN review · Curate in main workspace; return and confirm selection

DO THIS IN NANOSCOPE

1. **Select the prepared recording**
   Use the recording selector to choose the source you want to review. Check its identity and finish preparation first.

2. **Inspect the study selection**
   Show the candidate table to check decisions before opening the main workspace. The main Gallery and Inspector provide detailed editing.

3. **Open the main workspace**
   Click Curate in main workspace.... The main app opens the matching prepared trace and available study events.

Keep in mind

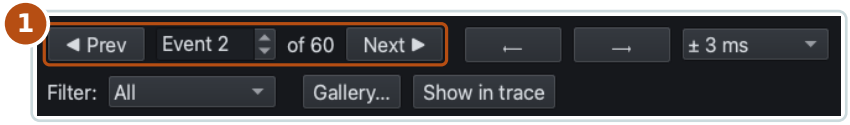
For a separate curated analysis, first save and copy the complete reference workspace. The prepared review trace is not the original open-pore current. Do not rerun detection if you only intend to review existing events.

You should see

The main workspace carries an attached study bar with Save curation to study and Return to study.

Accept, reject or correct individual events

Optional curated branch · work in a separate copy; counts may change



Reference capture, 21 Sep 2026. Choose the event and context; use Prev / Next to inspect each detection.



Reference capture, 21 Sep 2026. Accept or reject the displayed event. Revert mask becomes available after editing its bounds.



Reference capture, 21 Sep 2026. A real cu_2h event. The shaded interval is the event mask; native measurements use the prepared trace.

DO THIS IN NANOSCOPE

1 Inspect the selected waveform
Use ◀ Prev and Next ▶ in Event Inspector to move through events. Compare the shaded interval with the waveform.

2 Accept or reject
Click Accept or Reject. The Accepted checkbox reflects the decision; the action does not advance automatically. A/R and arrow keys offer the same review/navigation.

3 Correct bounds if necessary
Drag an event-mask edge and wait for remeasurement. Revert mask restores original bounds when available. Main Edit → Undo reverses supported edits.

Keep in mind

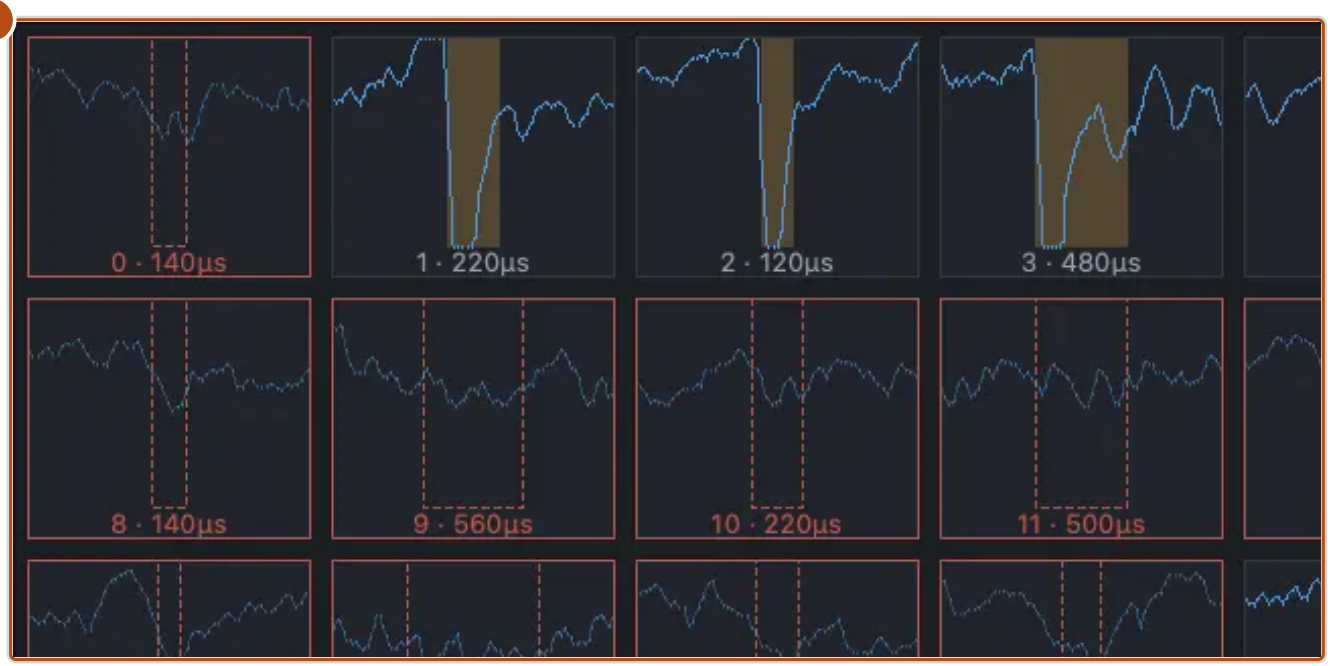
Open Event Inspector from the main Analysis menu after Curate in main workspace.... The study preview itself does not edit boundaries. Reconciled tables retain their source-editing restrictions.

You should see

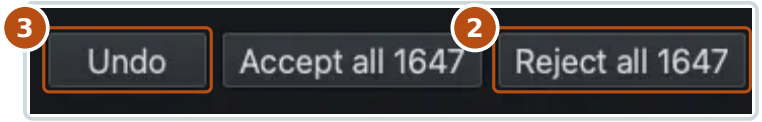
The main event table contains the intended acceptance and bounds. Next use Save curation to study and Return to study.

Review waveforms in the Event Gallery

Unified incubation workflow | NanoScope 40.15.59



Current Zn Gallery: cropped example tiles; remaining tiles are outside this crop.



Current Gallery bulk controls and Undo; read the full filter count.

DO THIS IN NANOSCOPE

1. **Compare the event tiles**
   From the main Analysis menu choose Event gallery.... Compare waveforms and open an event in the Inspector for precise boundary edits.

2. **Check the scope before a bulk decision**
   The all-events acceptance controls act on the current filter across pages. Read the displayed count. Use Accept selected or Reject selected for picked tiles instead.

3. **Undo an unintended edit**
   The Gallery Undo button uses the same undo history as the main workspace. Check the resulting event decisions after undoing.

Keep in mind

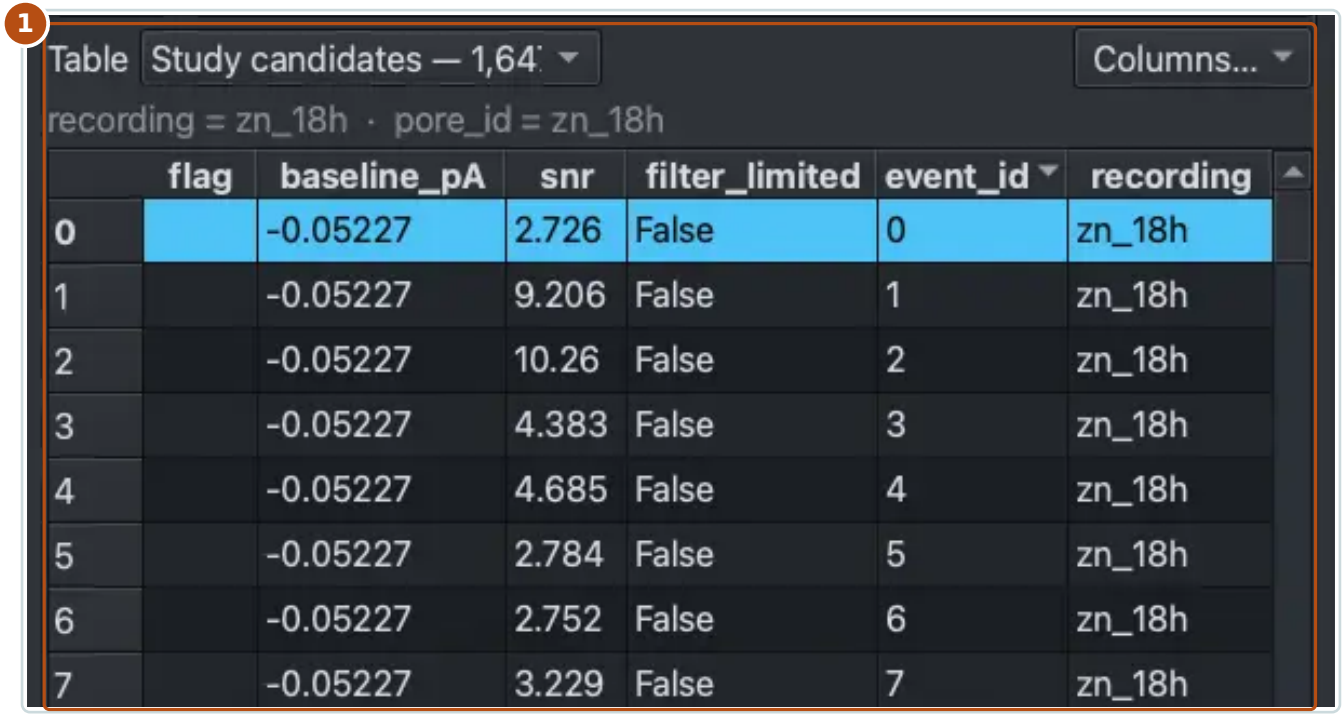
The screenshot is a cropped current-build view of saved Zn reference events. These controls are shared by all profiles. Reference reproduction retains its original acceptance decisions.

You should see

After Gallery edits, return to the main workspace and use Save curation to study before returning and confirming.

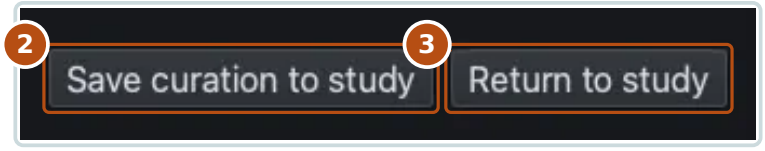
Save event edits back to the study

Unified incubation workflow | NanoScope 40.15.59



	flag	baseline_pA	snr	filter_limited	event_id	recording
0		-0.05227	2.726	False	0	zn_18h
1		-0.05227	9.206	False	1	zn_18h
2		-0.05227	10.26	False	2	zn_18h
3		-0.05227	4.383	False	3	zn_18h
4		-0.05227	4.685	False	4	zn_18h
5		-0.05227	2.784	False	5	zn_18h
6		-0.05227	2.752	False	6	zn_18h
7		-0.05227	3.229	False	7	zn_18h

Current event table, cropped to its identity and first rows.



Current study bar: save the curation before returning.

DO THIS IN NANOSCOPE

1. **Check the active events**
   Verify that the event table belongs to the recording you opened from the study. Review acceptance and edited bounds.

2. **Commit this curation**
   Click Save curation to study. Wait for successful completion; the study recalculates its measurements from its own raw and prepared signals.

3. **Return to the study**
   Click Return to study. Inspect the imported decisions and confirm the selection before recomputing affected analyses.

Keep in mind

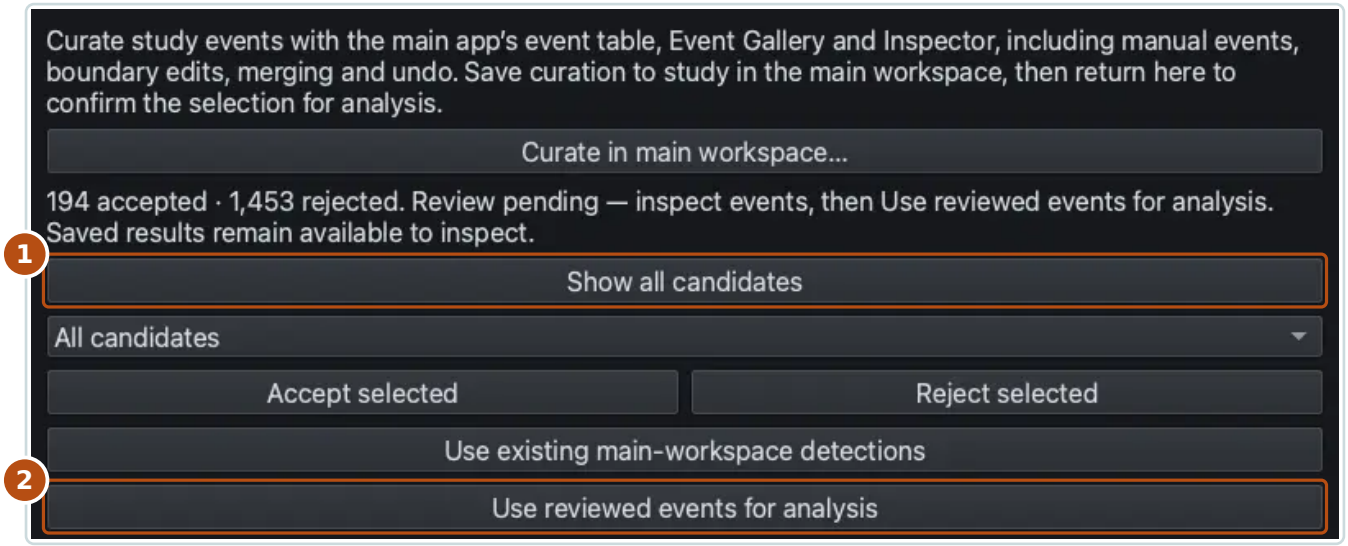
Save Project retains the main analysis session; it does not replace Save curation to study. Later event edits require another save. A failed or stale save is not a successful transfer.

You should see

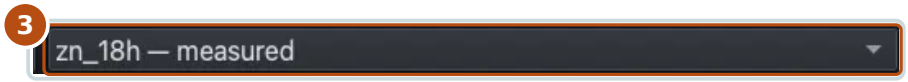
Acceptance and bounds are committed to the attached study recording. Dependent results must be recomputed after selection changes.

Confirm the reviewed selection

Unified incubation workflow | NanoScope 40.15.59



Current build 40.15.59. ZN review · Curate in main workspace; return and confirm selection



Current recording selector, cropped from the study toolbar.

DO THIS IN NANOSCOPE

1. **Inspect the saved events**
   Click Show all candidates. Check several edited events, their bounds, acceptance and recording identity.

2. **Approve this selection**
   Click Use reviewed events for analysis. Repeat for each included recording whose current selection is pending.

3. **Repeat for the other recordings**
   Select the next recording and complete its preparation, review and confirmation before pooled analysis.

Keep in mind

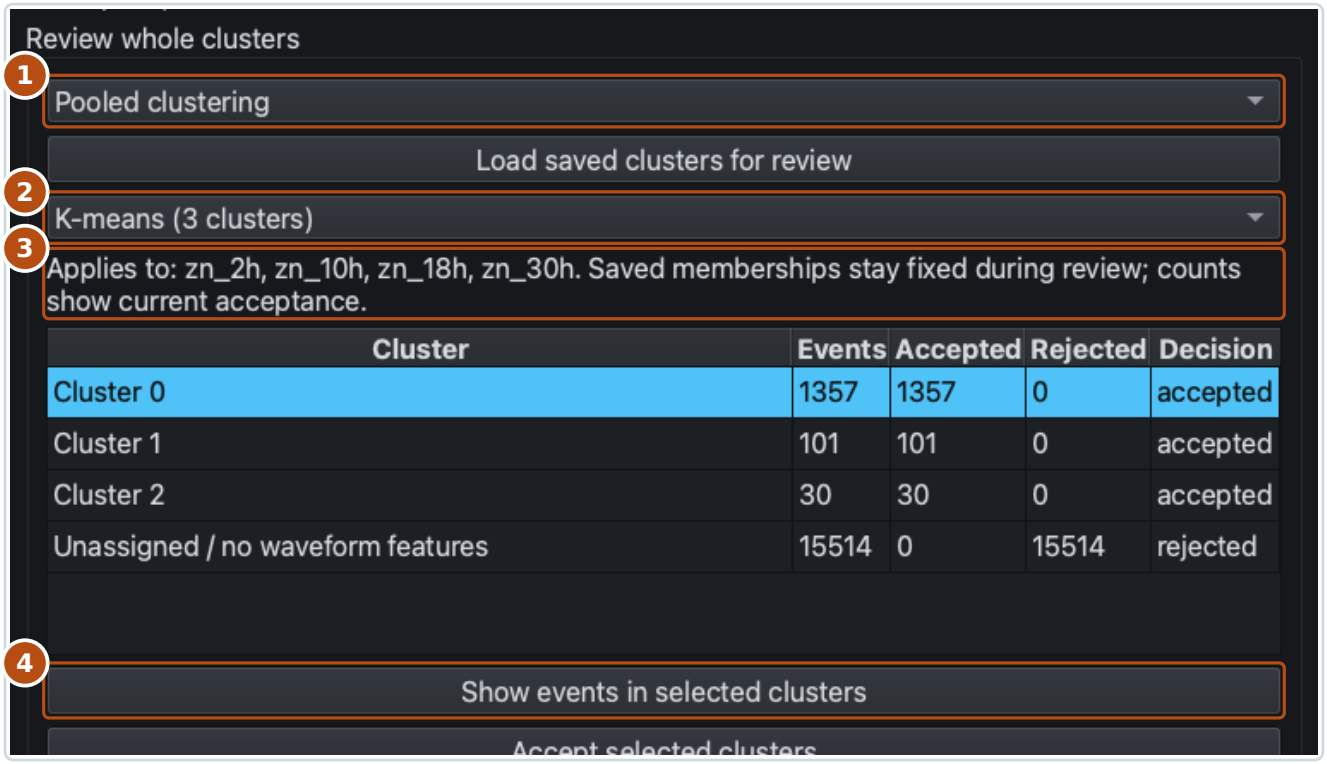
Confirmation approves the selection; it does not assert that every event was individually inspected. Automatic passes_gate remains distinct from accepted decisions. Reset review to automatic gate intentionally discards the current selection.

You should see

Recompute comparisons, features and fits from the final selection, then regenerate affected figures.

Inspect and select complete clusters

Optional curated branch · work in a separate copy; counts may change



Reference capture, 21 Sep 2026. Controls · scroll down

DO THIS IN NANOSCOPE

1 Choose the review source
First confirm individual review and run Z13 to create provisional groups (also Z14 for mixture review). At the TOP of 5 Features / populations, find the review-source selector.

2 Choose the saved grouping
Choose Pooled clustering or Selected-recording mixture. For older saved results click Load saved clusters for review. Select K-means (3 clusters), HDBSCAN or Mixture populations as appropriate.

3 Check the recording domain
Read Applies to and the Accepted/Rejected counts. Pooled groups cover their listed recordings; a selected-recording mixture covers only its target.

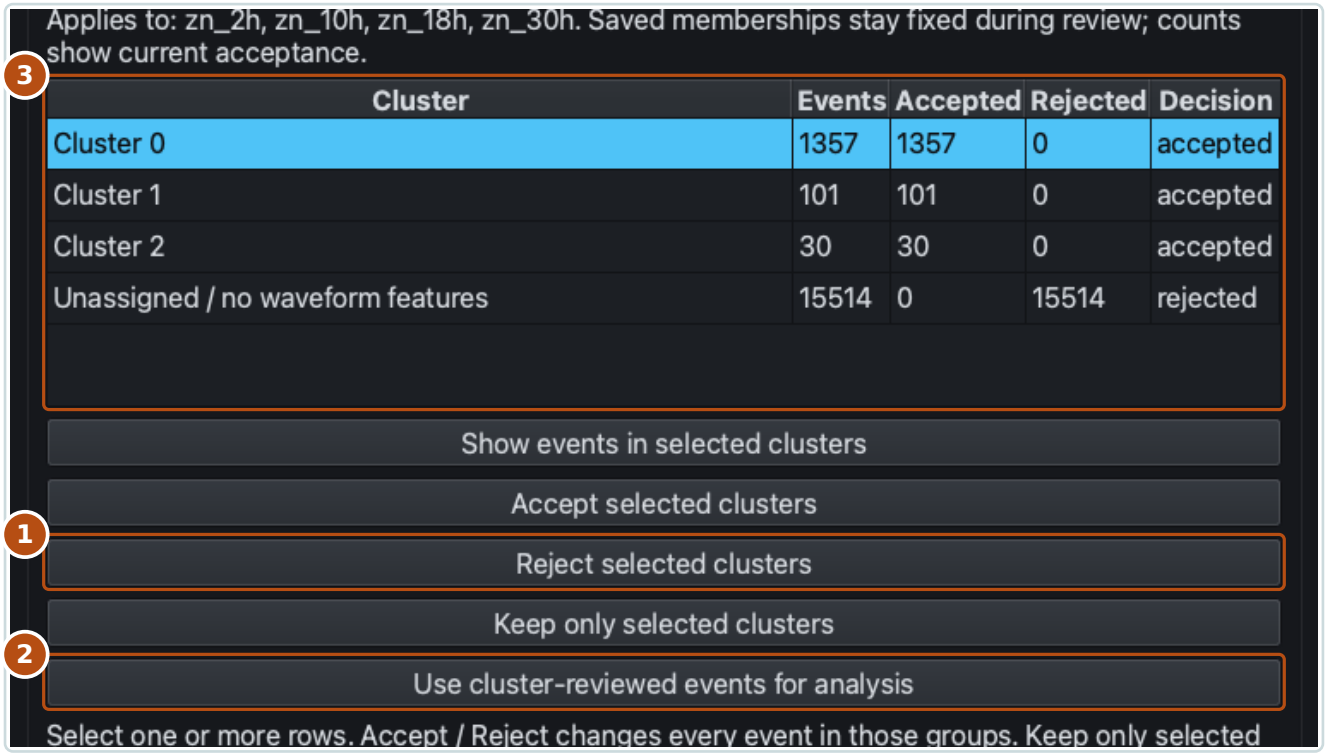
4 Inspect before deciding
Select one or several cluster rows and click Show events in selected clusters. Inspect their source events before accepting or rejecting the group.

Keep in mind
Unassigned / no waveform features is an explicit bucket. HDBSCAN Noise (-1) is a separate label, not an automatic reject. A cluster number means only this method and saved run; it is not a molecular identity.

You should see
You know which recordings and events the selected cluster rows control.

Apply cluster decisions and rerun the final analysis

Optional curated branch · work in a separate copy; counts may change



Reference capture, 21 Sep 2026. Controls · scroll down

DO THIS IN NANOSCOPE

1. **Apply the group decision**
   Accept selected clusters or Reject selected clusters affects only the chosen groups. Keep only selected clusters accepts those groups and rejects all other events in the displayed domain, including unassigned unless selected.

2. **Make exceptions, then confirm**
   Individual Accept/Reject on 3 Detect / review edits the same selection; the last action wins. When satisfied, click Use cluster-reviewed events for analysis.

3. **Inspect the final counts**
   Check Accepted and Rejected in the cluster table. Events outside the displayed recording domain remain unchanged. Confirm any other pending recording on 3 Detect / review.

Keep in mind

NEXT: rerun Z12 comparisons, Z13 feature extraction/clustering and Z14 mixture; regenerate Z16 figures and save. Saved memberships remain a review aid during acceptance edits, but new detection, measurement or main import retires them. Labels can change after refitting.

You should see

Final results describe the confirmed curated selection; counts need not match the reference.

A01 / Input files and model

Select these originals; do not substitute historical prepared CSVs.

Tag	Exact ABF filename
zn_2h	100 pM_Zn_2 h_Insulin_16 nm_50 percent 35k PEG_0000_-200 mV_Conductive_Good_0-4 min_Lab view corrrected.abf
zn_10h	100 pM_Zn_10 h_Insulin_18 nm_50 percent 35k PEG__0021_-200 mV_Conductive_ Very Good_Process_Lab view corrected.abf
zn_18h	100 pM_Zn_18 h_Insulin_18 nm_50 percent 35k PEG__0005_-200 mV_Conductive_Very good_Process_Lab view corrected_118 s.abf
zn_30h	100 pM_Zn_30 h_Insulin_20 nm_50 percent 35k PEG_0008_-200 mV_Conductive_Good_Process_225s.abf

Input folder: /Users/abrahamcampos/Desktop/Zn2IonCondition/

Model: /Users/abrahamcampos/Desktop/Zn2IonCondition/CSVData/Zn2student_zn_30h.keras

Notebook scope: Zn2Insulin_DataPreprocessing.ipynb and Zn2Insulin_StudentInference.ipynb. Training replacement models, acquisition-file corrections before export, and the separate calibrated CUSUM notebook are outside this reproduction.

R2h / Recording recipe · zn_2h

Use with Z03–Z11; all common settings remain mandatory.

Control / checkpoint	Value
Exact source	100 pM_Zn_2 h_Insulin_16 nm_50 percent 35k PEG_0000_-200 mV_Conductive_Good_0-4 min_Lab view corrrected.abf
Recording tag / Physical pore ID	zn_2h / zn_2h
Incubation / Pore diameter	2 h / 16 nm
Concentration / Voltage	100 pM / -200 mV
Channel / Scaling / Orientation	1 (IN11, pA) / Notebook float32 convention / Downward in source
Full samples / Duration	11,943,488 / 238.86976 s
Mask threshold / Dilation	3 MAD sigma / 0.2 ms
Candidates / Accepted	408 / 108
Model-uncovered tail	576 samples / 11.52 ms

Apply source details → prepare → inspect → infer with the supplied model → find candidates → measure/gate → inspect/confirm.

Shared source controls: added channel 0; pA override off; keep 0 to 0(end); exclusions blank; Reflected median edges; Supplementary off.

R10h / Recording recipe · zn_10h

Use with Z03–Z11; all common settings remain mandatory.

Control / checkpoint	Value
Exact source	100 pM_Zn_10 h_Insulin_18 nm_50 percent 35k PEG__0021_-200 mV_Conductive_ Very Good_Process_Lab view corrected.abf
Recording tag / Physical pore ID	zn_10h / zn_10h
Incubation / Pore diameter	10 h / 18 nm
Concentration / Voltage	100 pM / -200 mV
Channel / Scaling / Orientation	1 (IN11, pA) / Notebook float32 convention / Downward in source
Full samples / Duration	10,219,519 / 204.39038 s
Mask threshold / Dilation	3.5 MAD sigma / 2 ms
Candidates / Accepted	3,943 / 110
Model-uncovered tail	1,023 samples / 20.46 ms

Apply source details → prepare → inspect → infer with the supplied model → find candidates → measure/gate → inspect/confirm.

Shared source controls: added channel 0; pA override off; keep 0 to 0(end); exclusions blank; Reflected median edges; Supplementary off.

R18h / Recording recipe · zn_18h

Use with Z03–Z11; all common settings remain mandatory.

Control / checkpoint	Value
Exact source	100 pM_Zn_18 h_Insulin_18 nm_50 percent 35k PEG__0005_-200 mV_Conductive_Very good_Process_Lab view corrected_118 s.abf
Recording tag / Physical pore ID	zn_18h / zn_18h
Incubation / Pore diameter	18 h / 18 nm
Concentration / Voltage	100 pM / -200 mV
Channel / Scaling / Orientation	1 (IN11, pA) / Notebook float32 convention / Downward in source
Full samples / Duration	5,923,839 / 118.47678 s
Mask threshold / Dilation	3 MAD sigma / 2 ms
Candidates / Accepted	1,647 / 194
Model-uncovered tail	1,023 samples / 20.46 ms

Apply source details → prepare → inspect → infer with the supplied model → find candidates → measure/gate → inspect/confirm.

Shared source controls: added channel 0; pA override off; keep 0 to 0(end); exclusions blank; Reflected median edges; Supplementary off.

R30h / Recording recipe · zn_30h

Use with Z03–Z11; all common settings remain mandatory.

Control / checkpoint	Value
Exact source	100 pM_Zn_30 h_Insulin_20 nm_50 percent 35k PEG_0008_-200 mV_Conductive_Good_Process_225s.abf
Recording tag / Physical pore ID	zn_30h / zn_30h
Incubation / Pore diameter	30 h / 20 nm
Concentration / Voltage	100 pM / -200 mV
Channel / Scaling / Orientation	1 (IN11, pA) / Notebook float32 convention / Downward in source
Full samples / Duration	11,269,119 / 225.38238 s
Mask threshold / Dilation	3 MAD sigma / 2 ms
Candidates / Accepted	11,004 / 1,076
Model-uncovered tail	1,023 samples / 20.46 ms

Apply source details → prepare → inspect → infer with the supplied model → find candidates → measure/gate → inspect/confirm.

Shared source controls: added channel 0; pA override off; keep 0 to 0(end); exclusions blank; Reflected median edges; Supplementary off.

A02 / Copyable feature lists

Paste the entire comma-separated line into the named GUI field.

Control / checkpoint	Value
Pooled clustering features (12)	log_dwell, depth_max_pA, depth_sust_pA, mean_pA, std_pA, min_dev_pA, skewness, kurtosis, area_pA_ms, t_peak_frac, f_sust, snr
Mixture features (13)	log_dwell, depth_max_pA, depth_sust_pA, mean_pA, std_pA, min_dev_pA, skewness, kurtosis, area_pA_ms, t_peak_frac, f_sust, snr, entropy

Pooled settings: seed 0; K-means 3; PCA variance 0.95; max PCs 10; HDBSCAN min samples 10; descriptive t-SNE on.

Mixture target zn_30h; PCA grid 2,3,4,5,6,7,8,9,10,11,12,13; K grid 1,2,3,4,5,6,7,8; folds 5; bootstrap 200; fit seed 0; bootstrap seed 0; minimum-events field 0.

Feature calculations use the whole prepared-trace median/MAD and global float64 raw median. Tails are clipped at 1st/99th percentiles, then median/IQR scaled. HDBSCAN minimum cluster size is max(20,N//100); reference t-SNE perplexity is 29. These are recorded method choices, not extra fields.

Paste-ready companion files in the manual bundle: Zn_pooled_features.txt and Zn_mixture_features.txt. Copy the entire line into its matching field.

A03 / Reference comparison checkpoints

Automatic acceptance from the raw ABFs; do not force a curated result to these values.

Recording	Rate (events/s)	95% interval	Median sustained/local	Median dwell (ms)
zn_2h	0.452129	0.368402–0.569348	0.063424	0.12
zn_10h	0.538186	0.450119–0.631145	0.084081	0.28
zn_18h	1.637452	1.316714–2.008832	0.067491	0.22
zn_30h	4.774109	4.144068–5.488450	0.060715	0.24

Fractional blockade is dimensionless; multiply by 100 for percent.

Rates and intervals describe within-recording uncertainty; they do not create independent pore replicates. Do not average the four rates as independent biological replicates.

A04 / Read the three amplitude definitions

The name of the saved column determines its meaning.

Column	Meaning
fracblock	Sustained depth / absolute time-local raw I0. Sustained depth = quiet reference minus the event median.
aggregation_fracblock_global	Peak depth / absolute whole-recording float32 raw median. Peak depth = quiet reference minus the within-event minimum.
f_sust_global	Sustained depth / the same global raw median.

Time-local I0 uses quiet raw-current block medians, smoothed and interpolated at onsets. The near-zero prepared baseline is not raw I0.

The gate-transfer sweep applies an extra floor to already accepted events. Native gate reference uses preparation-QC noise / global I0; detector event-excluded noise is separately recorded.

Pair ratios use the nearest sweep-grid value to the pair’s larger native floor. KS comparisons use the exact largest native floor among recordings with ≥50 accepted events and need ≥20 above-floor events in each arm. Common-floor KDE uses the largest native floor and sustained/global amplitude.

Event start_idx and end_idx are inclusive. Dwell = (end_idx-start_idx+1)/sample_rate; exported end_s is exclusive.

Time-half diagnostics use the recording clock midpoint; the onset-median split is retained only as a legacy comparison.

A05 / The known 10 h numerical difference

Keep the new raw-ABF calculation; do not tune to recreate two historical rejects.

Control / checkpoint	Value
Fresh raw reference	3,943 candidates / 110 accepted
Historical prepared CSV	3,945 candidates / 110 accepted
Accepted sample bounds and dwell	All 110 are identical
Historical vs current preparation	230,078 / 10,219,519 baseline/prepared samples differ; max 0.333267 pA
Accepted sustained-depth difference	Max 0.064265 pA (0.5345%); median -0.000636 pA
Accepted sustained-fraction difference	Max 0.0003973; local/global I0 unchanged

Stable direct NumPy median checks agreed with the current baseline at all checked positions; the historical baseline disagreed at all 128 sampled disputed positions.

28 rejected intervals shift; five historical rejects disappear and three new rejects appear. No shared-boundary gate decision flips.

Raw current matches for all four sources. Baseline/prepared samples match the historical preparation exactly at 2, 18 and 30 h.

The original prepared files are useful for exact historical replay through the automated shortcut, but this guide reproduces from raw ABFs.

A06 / Figure inventory · recording inspection

Five figures for each of zn_2h, zn_10h, zn_18h, zn_30h = 20.

File stem	Contents
gallery50_<tag>	Up to 50 accepted events ranked by peak depth / quiet MAD; onset separation >20 ms.
gallery_sustained_<tag>	Up to 50 accepted events ranked by sustained depth / quiet MAD; onset separation >20 ms.
typical_events_<tag>	Up to four accepted events near the median log-dwell/peak-depth coordinates; onset separation >50 ms.
trace_context_<tag>	Whole trace and deepest excursion; raw and prepared current.
single_recording_candidates_<tag>	All fired candidates; depth/dwell histograms and scatter.

50-event ladders are ranked examples, not random samples. Their local display recentering (quiet samples within ±20 ms) does not replace scientific amplitude columns. Typical-event panels use the prepared current without that local recentering.

The all-candidate summary follows the notebook’s separate full-trace-median peak depth and rounding: 40 histogram bins, depth rounded to 0.1 pA and dwell to 0.001 ms.

A07 / Figure inventory · comparisons and populations

Seven comparisons + two pooled views + three mixture figures = 12.

File stem	Contents
incubation_sustained_local	Dwell and sustained/local distributions
aggregation_peak_global	Dwell and peak/global distributions
kde_sustained_global	Sustained/global distributions
size_matched_18nm_sustained_local	10 h vs 18 h: separate physical pores, same nominal diameter
incubation_rates_and_medians	Rates, time-block intervals, amplitude/dwell medians
gate_transfer_sweep	Rates above floors 0.02–0.20
common_floor_kde	Sustained/global distributions above the common floor
pooled_pca / pooled_tsne	Views colored by K-means, HDBSCAN, time and log-dwell
quantitative_appendix_selection_fractions_waveforms	30 h selection, weights, hard fractions and median waveforms
population_examples_zn_30h	Up to four representative events per component
population_onsets_zn_30h	Assigned events: upper minimum markers and lower onset timeline

A08 / Export a multi-panel publication figure

Optional main-app Figure Library route after study figures are final.

1. In each study figure editor click Save to Figure Library… and enter a descriptive name. Save at least two entries. In the main window use File → Save Project… to persist the library.

2. Open Analysis → Figure Library… → Compose multi-panel figure…. Check Use for 2–12 figures. Set panel labels and Columns. Use Rows/Cols spans, Move up/down, Panel spacing and panel-label appearance to arrange them.

3. Keep individual axes/legends unless the selected panels support compatible shared controls; a study figure may already contain several panels. Click Update preview and inspect.

4. Click Save to Figure Library, save the main project again, then Export saved figure… or Save figure bundle….

5. Revise composition… creates a new saved revision. Library entries are frozen snapshots; later study curation or styling does not update them. Save new entries after a change.

Custom event captions follow the same event and trace. Changed events use automatic captions until restyled; older captions without identity must be re-entered.

A09 / Rerun from the first changed stage

A changed field is a proposed setting until that operation completes.

Changed input	First operation to repeat	Then
Source details	Apply recording details → Prepare selected recording	Inference → candidates → measurement → confirm → downstream
Preparation	Prepare selected recording	Inference → candidates → measurement → confirm → downstream
Model / normalization / stride / coverage	Run model inference	Candidates → measurement → confirm → downstream
Probability thresholds / merge / duration	Find candidate events	Measurement → confirm → downstream
Gate / I0 measurement	Measure and gate candidates	Confirm → comparisons/features/fits → figures
Acceptance or main-workspace events	For native edits: Save curation to study, then confirm review	Comparisons → feature extraction → fits → figures
Cluster acceptance	Use cluster-reviewed events for analysis	Comparisons → feature extraction → final fits → figures
Comparison settings	Compute recording comparisons	Comparison figures
Pooled settings	Run pooled exploratory clustering	Pooled figures
Mixture settings/target	Fit selected-recording mixture	Mixture figures
Appearance only	Apply appearance → Save styling to study	No scientific recalculation

If an upstream edit clears figures, finish scientific operations and regenerate all affected groups. Prior files remain in timestamped runs folders.

Pending review: confirm each included recording. Pending parameters: Restore completed settings or deliberately rerun; a draft on another recording can also block downstream work.

Main edits absent: use Save curation to study on the main study bar, Return to study, inspect and confirm. Newer workspace revision: reopen the latest saved folder; keep one editing session. Missing scientific file: restore the complete workspace from an intact copy.

SHARED / Shared workflow for Zn, Cu, acid and future studies

All studies open through Analysis > Incubation study.... General, Zn and Cu use the editable staged workspace. Acid retains its calibrated CUSUM method and frozen-family definitions. Changing the entry point does not make their scientific detectors interchangeable.

For a future experiment choose General incubation, name the sample and conditions, and enter the actual time, pore identity, concentration, voltage, channel and retained intervals. Choose and document preparation, detection and measurement settings appropriate to that study. A model is optional; selecting a profile does not validate a new experiment.

The main event table, Gallery and Inspector are shared. Save curation to study commits a snapshot. Return, confirm the selection and recompute dependent analyses. Keep automatic and human decisions distinguishable, and do not treat clusters as molecular identities.

Advanced: original Zn reference recipe... is the fixed automated reference route. Its automated run uses the fixed recipe. For curated analysis, save events to the staged study, confirm the selection and run that study's downstream stages.